

diffctx: Budgeted Typed-Graph Retrieval for Diff-Aware Code Context Selection

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<https://github.com/nikolay-e/diffctx>

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Abstract

Retrieving optimal context for Large Language Models to understand code changes is a critical challenge in automated software engineering. Current approaches rely on naive windowing or purely lexical retrieval, missing complex structural dependencies. We formulate diff-aware code context selection as **budgeted utility maximization** over multi-resolution fragments on a typed dependency graph. The deployed system maximizes a monotone submodular concept-coverage objective (concave saturation over per-concept relevance derived from the diff) with lazy-greedy budgeted selection and engineering heuristics; a modular cost-penalized (Boltzmann) ranking variant is available as an opt-in ablation, and we identify clean analyzable variants whose approximation guarantees are known. Two structural components ground the framework: (i) a partition matroid over multi-resolution fragments enforcing structural non-redundancy, and (ii) a pluggable relevance signal $\hat{w}(f, \Delta)$ instantiated via three families—bounded ego-network expansion (the deployed default), Personalized PageRank, and BM25 lexical retrieval. The framework is evaluated by file-level recall against annotated golden contexts at a fixed token budget across three multi-language software-engineering benchmarks (1500 instances total; consolidated rerun, v4-calibrated). At an 8000-token budget under bounded ego-network scoring, diffctx achieves pooled file recall 0.919 [95% CI 0.908, 0.929], with per-benchmark recall 0.998 on SWE-bench Verified, 0.942 on PolyBench-500, and 0.816 on ContextBench Verified. Decomposing recall into changed-file retention and genuine context retrieval reveals that only ContextBench Verified carries gold files beyond the input diff; on that retrieval subset (213 instances), the internal BM25 lexical mode is the strongest single signal (0.402 nontrivial recall) versus 0.352 for the deployed graph-lexical mixture and 0.244 for PPR; the signals are complementary, and a runnable score-free union of the two modes’ selections reaches 0.473 nontrivial and 0.870 headline recall at a measured deduplicated cost of 6.1k mean tokens — within the same budget for 91.5% of instances — making calibrated in-scorer combination the highest-leverage open item. A within-framework scoring-mode ablation (EGO vs. PPR vs. internal BM25), operational ablations (lexical component, relatedness bonus, adaptive stopping, Boltzmann objective), trivial floors (changed-files-only, random packing), and budget-grid sweeps over $\{0, 8k, 16k, 32k, 64k, 128k, -1\}$ are reported in Section 6. Against same-budget external baselines, paired both-OK deltas all favor diffctx: +0.371 [+0.350, +0.392] over whole-file BM25 packing, +0.423 [+0.399, +0.447] over Aider repo-map (fair-input), and +0.410 [+0.385, +0.434] over the Aider oracle-mentioned upper bound (permutation $p = 10^{-5}$ throughout); unlike the previous snapshot, external BM25 no longer overtakes diffctx at any measured budget.

1 Introduction

Large Language Models have demonstrated remarkable capabilities in code understanding tasks, yet their effectiveness depends critically on the context provided within limited token budgets. When reviewing code changes, developers and AI systems alike must determine which surrounding code fragments are necessary to understand the implications of a diff. Too little context leads to misunderstanding; too much context wastes tokens and introduces noise.

Current approaches to context selection suffer from fundamental limitations. Fixed-window methods (e.g., ± 10 lines around changes) ignore semantic structure and miss distant dependencies. Whole-file inclusion is wasteful and scales poorly. Lexical retrieval (BM25, embeddings) can miss structural relationships that are often important for code comprehension [15, 27]. None provide principled stopping criteria or approximation guarantees.

We address this gap by formalizing diff context selection as **budgeted submodular maximization** over Code Property Graphs. Our key insight is that “understanding a change” can be operationalized as covering explanatory concepts—definitions, call targets, type signatures, impacted tests—that propagate through structural dependencies with diminishing returns.

Task scope: post-hoc diff understanding. We address *post-hoc diff understanding*: a patch Δ is given as input, and the task is to retrieve a compact set of explanatory code fragments needed to review or reason about that patch within a token budget. This is distinct from *pre-resolution issue resolution* (the standard SWE-bench setup), where an agent must produce Δ from a problem statement without seeing the gold patch. The benchmarks in Section 5 are repurposed from issue-resolution datasets: we use their gold patches as known diffs against which file-level retrieval is scored, never as a target an agent must generate. The patch is on the input side; the recall metric measures whether the retrieved context covers the files the patch touches and the dependency neighborhood needed to interpret them.

Contributions.

1. A formal problem statement casting context selection as budgeted utility maximization over a partition matroid on multi-resolution fragments. We do not claim a new approximation algorithm; we use known optimization structure to make context selection explicit, analyzable, and extensible. An “algorithm $\times$ constraint $\times$ guarantee” table (Table 1) separates the deployed heuristic from analyzable variants; the deployed objective is the submodular concept-coverage form of Section 4.5
2. A typed-edge dependency-graph framework with per-category extraction and treatment (e.g., hub-suppression exemption), generalizing flat code-graph approaches
3. A pluggable relevance-scoring abstraction $\hat{w}(f, \Delta)$ where three algorithm families (bounded ego-network expansion, Personalized PageRank, BM25) instantiate the abstract signal under a single retrieval framework
4. An evaluation protocol using file-level recall against annotated golden contexts at a fixed token budget on multi-language software-engineering benchmarks, with paired-bootstrap statistical comparison against external baselines (BM25 over patch identifiers; Aider repo-map)
5. A reference implementation with structural parsing for ≈ 40 source and configuration formats and 36 dedicated semantic edge-builder modules covering 45+ languages/formats, plus a six-builder config-edge family (Appendix D). Eight languages (Python, JavaScript, TypeScript, Java, Go, Rust, C, C++) appear in the empirical evaluation with $n \geq 10$ instances; the rest are supported but unvalidated at scale (Table 16)

2 Related Work

Program Dependence and Slicing. Program slicing [30] computes the subset of statements affecting a variable, forming the foundation for dependency-based code understanding. Program Dependence Graphs (PDGs) [7] and their interprocedural extensions [12] capture control and data dependencies, enabling precise impact analysis. Our work extends these ideas to context selection for LLMs, treating dependency graphs as the substrate for relevance propagation.

Developer Navigation and Information Foraging. Studies of how developers navigate code [15, 27] reveal that structural relationships (calls, definitions) dominate over lexical search. This informs our edge weighting scheme, prioritizing structural signals while retaining lexical features as fallback.

Submodular Optimization. Monotone submodular maximization admits efficient greedy approximations [23], with lazy evaluation [21] and stochastic variants [22] improving scalability. We apply this framework to context selection, modeling coverage of explanatory concepts as a submodular function.

Retrieval for Code Understanding. Recent work explores retrieval-augmented generation for code tasks, typically using embedding similarity or BM25 over code chunks. Our approach differs by grounding retrieval in explicit dependency structure rather than purely lexical or semantic similarity, and by providing a principled utility function for budget allocation.

Repository-Level Context Tools. Aider [9] proposes a related repository-mapping approach using Personalized PageRank over tree-sitter-extracted definition–reference graphs, with binary-search packing under a token budget. Our framework differs in three respects: (i) we formalize selection as budgeted utility maximization over a partition matroid with multi-resolution fragments, identify analyzable variants of the formulation whose approximation guarantees are known, and clearly separate the deployed heuristic from those analyzable variants, rather than presenting a heuristic packing without the formulation; (ii) we introduce a typed edge taxonomy with per-category treatment (e.g., hub-suppression exemption for semantic, structural, and test edges), and (iii) we treat scoring as a pluggable component, comparing PPR, ego-network expansion, and lexical baselines empirically. Sourcegraph Cody [10] formalizes code retrieval as a multi-stage RecSys pipeline with complementary retrievers (keyword, embedding, graph), but does not commit to a formal optimization framework. RepoMapper [6] replicates Aider’s PageRank approach as an MCP server.

Concurrent Work on Submodular Code Context. Concurrent work by Arafat [2] explores submodular packing for citation-grounded comprehension on conversational queries over Python repositories, combining BM25, dense embeddings, and graph expansion via import relations; their evaluation reports completeness improvements over plain greedy via submodular packing. Our work differs in three concrete respects: (i) diff-aware setting with explicit core edit set E_0 and budget reservation, versus their conversational query setting; (ii) typed edge taxonomy with ten categories and per-category treatment, versus their flat graph; (iii) explicit matroid–knapsack analysis with temperature reweighting, versus their direct submodular maximization over a single retrieved candidate set. The two works are complementary and together support the broader case for submodular formulations in code context selection.

Impact Analysis and Co-change. ATHENA [32] combines dependency graphs with transformer embeddings for impact analysis on bug-fix prediction; Ripple [31] extends this with LLM reasoning over evolutionary coupling. These works target predicting which files *will be*

changed given a seed, an adjacent task to context selection. DraCo [5] replaces random walks with static dataflow analysis for retrieval, demonstrating that explicit structural reasoning can outperform learned propagation in code-specific settings. We share their structural emphasis but differ in objective (context for understanding rather than prediction of next change) and formal framework.

3 Problem Formulation

3.1 Inputs and Definitions

Given repository states $V_{\text{before}}, V_{\text{after}}$ with diff $\Delta = \text{diff}(V_{\text{before}} \rightarrow V_{\text{after}})$, token budget B , and (during evaluation) an annotated golden set $G \subseteq F$ of fragments deemed relevant for understanding Δ , we seek a fragment set $C^* \subseteq F$ that maximizes weighted recall against G —approximated at inference by a hand-constructed relevance estimator $\hat{w}(f, \Delta)$ (Section 4.2)—subject to a partition matroid structure on F and a token budget $\text{cost}(C) \leq B$.

Fragment Universe F . The set of candidate fragments is defined as

$$F = \{(\text{source}, \text{start}, \text{end})\}, \quad (1)$$

contiguous, syntactically or semantically delimited regions of the sources, each carrying a stable identity. The notion is domain-agnostic: in source code, fragments are named declarations and their bodies (functions, types, modules, constants, rules); in prose, sections, paragraphs, or list items; in structured data (YAML, JSON, XML), keyed entries and array elements; in logs, individual events. Extraction is deferred to a domain-specific parser (tree-sitter for code, structural splitter for markup, regex-delimited for unstructured text), with indentation- or whitespace-based fallback when parsing fails. Fragments may also exist at multiple granularities for the same underlying unit—e.g., a full function body and a signature-only abridgement, or a full section and its heading—which introduces structural redundancy handled below.

Non-Redundancy via Partition Matroid $M = (F, \mathcal{I})$. Fragments in F may be mutually redundant: one may be nested inside another, multiple variants may exist at different granularities, or the same content may recur across documents. We define the redundancy relation $\sim$ on F *narrowly*: two fragments are equivalent iff they are alternative resolutions of the same underlying unit (e.g. a full function body and its signature-only abridgement) or byte-identical duplicated content. So restricted, $\sim$ is an equivalence relation as given—no closure is needed—and partitions F into classes $\mathcal{P} = \{P_u\}$, of which at most one representative may be selected:

$$\mathcal{I} = \{C \subseteq F : |C \cap P_u| \leq 1 \text{ for all } u\} \quad (2)$$

We deliberately do *not* fold general range overlap into $\sim$: taking the transitive closure of “ranges overlap” would chain nested and staggered fragments into large merged classes (under which the core set E_0 —hunk fragments plus their enclosing containers—becomes infeasible, since a container and its child overlap). Overlap is instead a second, pairwise constraint on the selected set: no two selected fragments may overlap (up to a deliberate one-line boundary tolerance for back-to-back fragments), and a candidate fully contained in an already-selected fragment is rejected. The implementation enforces exactly this: a per-source sorted interval index performs pairwise overlap and containment tests of each candidate against the current selection, plus a signature-vs-full deduplication step; it builds no connected components and no explicit equivalence classes. The resulting feasible family (partition constraint on the narrowed classes $\cap$ pairwise non-overlap) is an independence system; the matroid structure holds exactly for the narrowed $\mathcal{P}$, and the pairwise overlap constraint is treated as part of feasibility, not of the matroid.

Cost Model (modular proxy).

$$\text{cost}(C) = \sum_{f \in C} |f| \quad (3)$$

where $|f|$ denotes the size of fragment f in consumer-appropriate units (tokens for LLM consumers, bytes for storage-bound channels, lines for human readers), and any per-fragment

metadata cost (source identifiers, position markers, delimiters) is folded into $|f|$ at extraction time. This is a *modular* approximation: the true assembled cost is bounded above by $\text{cost}(C)$, with overlap deduplication across nested fragments and shared per-source headers acting as a free reduction at materialization time. Modeling these second-order savings would render cost submodular and complicate optimization without changing feasibility. For LLM consumers, practical budgets range from 5k to 20k tokens, bounded by context window limits and the “lost in the middle” effect [19].

Knapsack Constraint. The cost model induces a knapsack constraint independent of the matroid:

$$\mathcal{K} = \{C \subseteq F : \text{cost}(C) \leq B\} \quad (4)$$

Feasible Region. Let $\mathcal{J}$ denote the pairwise non-overlap family introduced above ($\mathcal{J} = \{C : \text{no two fragments in } C \text{ overlap beyond the one-line boundary tolerance}\}$, an independence system but not a matroid). The deployed feasible region is the triple intersection

$$\mathcal{F} = \mathcal{I} \cap \mathcal{J} \cap \mathcal{K} = \{C \subseteq F : |C \cap P_u| \leq 1 \text{ for all } u, C \in \mathcal{J}, \text{ and } \text{cost}(C) \leq B\} \quad (5)$$

We separate the three deliberately: the matroid $\mathcal{I}$ and the overlap family $\mathcal{J}$ are hard structural constraints, while the knapsack is a numerical resource constraint; the analyzable variants below drop $\mathcal{J}$ (their guarantees are stated on $\mathcal{I}$, $\mathcal{K}$, or $\mathcal{I} \cap \mathcal{K}$ alone). The literature provides several reference results for clean analyzable variants:

- **Modular function under a matroid:** greedy on the cost-blind weighted matroid problem returns the exact optimum (Edmonds; Rado–Edmonds matroid theorem), simply “one best representative per class”;
- **Monotone submodular under a cardinality constraint $|C| \leq k$:** greedy attains $(1 - 1/e)$ [23];
- **Monotone submodular under a general matroid (greedy):** $1/2$ [8] (Fisher–Nemhauser–Wolsey, the matroid theorem); $(1 - 1/e)$ via continuous greedy [3];
- **Monotone submodular under a knapsack:** $\frac{1}{2}(1 - 1/e)$ via density-greedy plus the arg max-singleton modification—proved for coverage in [14] and for general monotone submodular objectives in [16, 17]; $(1 - 1/e)$ via partial enumeration [29];
- **Monotone submodular under matroid $\cap$ knapsack:** $(1 - 1/e - \varepsilon)$ via continuous greedy with contention resolution [4].

The deployed system is a heuristic and we do not invoke any of the above guarantees directly on it. Table 1 maps each variant the framework supports to its objective form, constraint, algorithm, and resulting claim.

Objective. The objective takes the same algebraic form in both modes but with different sources for the per-fragment weight $w(f, \Delta) \geq 0$:

$$\max_{C \subseteq F} \mathcal{U}(C) = \sum_{f \in C} w(f, \Delta) \quad \text{s.t. } C \in \mathcal{F}. \quad (6)$$

Evaluation mode (golden set G available): $w(f) = \mathbb{1}[f \in G] \cdot \omega_f$, so the sum collapses to weighted recall against G :

$$\mathcal{U}_{\text{eval}}(C) = \sum_{f \in C \cap G} \omega_f. \quad (7)$$

Table 1: Algorithm $\times$ constraint $\times$ guarantee map. The deployed default is a heuristic instantiating the submodular concept-coverage objective of Section 4.5; analyzable variants of the framework admit the listed guarantees on their respective surrogate problems. The modular Boltzmann-ranked variant is opt-in.

Variant	Objective	Constraint	Algorithm	Claim
Deployed default (this paper)	monotone submodular concept coverage (Section 4.5) + adaptive stopping + rescue	budget $\cap$ partition matroid $\cap$ pairwise non-overlap ($\mathcal{I} \cap \mathcal{J} \cap \mathcal{K}$)	lazy density-greedy with heuristics	empirical + additive stopping certificate (Prop. 1)
Modular cost-blind variant	modular	partition matroid	best representative per class	exact (Edmonds)
Modular + knapsack, arg max-singleton modification	modular	knapsack	modified density-greedy	$\frac{1}{2}$ (classical linear-knapsack bound; the modular case of [14])
Clean submodular variant (Section 4.5, no stopping/bonuses)	monotone submodular	knapsack	modified density-greedy	$\frac{1}{2}(1-1/e)$ [14]; $(1-1/e)$ w/ enumeration [29]
Clean submodular variant, matroid only	monotone submodular	partition matroid	greedy; continuous greedy	$1/2$ greedy; $(1-1/e)$ continuous [23, 3]

Inference mode (no golden set): $w(f, \Delta) = \hat{w}(f, \Delta)$, a scoring function (Section 4.2) that estimates per-fragment relevance to Δ without access to G .

The modular form above is the analyzable simplification used in the guarantee discussion; in deployment the per-fragment relevance feeds the submodular concept-coverage aggregate of Section 4.5. Precision is controlled implicitly via the budget B , which bounds $|C'|$. Asymmetric error cost—false negatives (missing a relevant fragment) degrade LLM understanding more severely than false positives—justifies the recall-centric formulation in both modes.

Modular vs. Submodular Instantiations. The framework supports both modular and submodular instantiations of $\mathcal{U}$ depending on how w is constructed. When w aggregates concept coverage through a nondecreasing saturation ϕ (Section 4.5; monotonicity of ϕ suffices for submodularity of the max-coverage form, concavity additionally shapes the saturation profile)—*the deployed objective in this work*— $\mathcal{U}$ is monotone submodular, and the *clean variant* of the modified density-greedy (knapsack constraint only, no adaptive stopping, no path-dependent bonus) carries the $\frac{1}{2}(1-1/e)$ guarantee [14, 16]; greedy under the partition matroid alone attains $1/2$ and continuous greedy attains $(1-1/e)$ [8, 3]. The *deployed configuration* adds the partition constraint, core packing, adaptive stopping, and bonuses on top of the knapsack, and is a heuristic: none of the listed constants transfer to it (Table 1). When w linearly weights fragments by relevance (the opt-in Boltzmann-ranked variant), $\mathcal{U}$ is modular: under a partition matroid alone the problem is exactly solvable by selecting the best representative per equivalence class (the trivial matroid optimum); under a knapsack the same modified density-greedy is $\frac{1}{2}$ -approximate on the modular-knapsack instance. We do not deploy continuous greedy. The optimization machinery is identical across modes; only the guarantee statement and the precise objective differ.

Pluggable Relevance Estimator. We treat $\hat{w}(f, \Delta)$ as a pluggable scoring component of the framework. In this work we instantiate it via three families: (i) *bounded ego-network expansion* with depth-decayed weights, the deployed default (Section 4.2.2); (ii) *Personalized PageRank* over a typed dependency graph (Section 4.2.1); (iii) *lexical similarity* via BM25 over fragment tokens (Section 4.2.3). The framework of this section is independent of which $\hat{w}$ is chosen; the analyzable variants of Table 1 apply uniformly to any non-negative scoring instantiation.

Cost-Penalized Surrogate (Lagrangian-Style). An optional cost-aware reweighting folds fragment cost into the per-fragment score:

$$\tilde{w}(f, \Delta) = w(f, \Delta) \cdot \exp(-\beta \cdot |f|), \quad (8)$$

yielding the surrogate objective $\tilde{\mathcal{U}}(C) = \sum_{f \in C} \tilde{w}(f, \Delta)$ over the partition matroid M . This is an exponential cost-penalized reweighting—a Lagrangian penalty in log-weight space, $\log \tilde{w} = \log w - \beta|f|$, not a Lagrangian relaxation of the budgeted objective itself: it solves a different penalized problem rather than the original hard-budget objective, and the post-hoc budget truncation that produces a feasible $C \in \mathcal{F}$ does not preserve the surrogate’s optimality. We use it as a budget-aware ranking signal during selection, not as an approximation algorithm for the original problem; we therefore do not claim that β -reweighting recovers an approximation guarantee for $\max \mathcal{U}$ subject to $\text{cost}(C) \leq B$. The parameter β is calibrated by binary search to hit a target expected cost on a validation set (Section 4.4); $\beta \rightarrow 0$ recovers pure relevance ranking, $\beta \rightarrow \infty$ collapses to cost minimization.

Core Edit Set: E_{raw} vs. E_{pack} . From diff Δ the implementation first collects a *raw* core set E_{raw} : for each hunk, the smallest covering fragment (else all overlapping fragments, else the nearest fragment), plus candidate enclosing syntactic containers. E_{raw} is in general *infeasible* as a set—a container and its child overlap, and multiple resolutions of one unit may co-occur. The set that actually precedes the greedy loop is the *packed* core $E_{\text{pack}} \subseteq E_{\text{raw}}$: one resolution per unit, containment conflicts rejected pairwise, oversized members downshifted to signature variants, packed under the reservation below. The contraction claim is therefore stated at E_{pack} : rather than assigning $w \rightarrow \infty$, the implementation optimizes subject to $E_{\text{pack}} \subseteq C$, i.e. greedy on the objective contracted at E_{pack} (contraction preserves monotone submodularity), with $E_{\text{pack}} \in \mathcal{F}$ feasible by construction. Symbol-level context makes relevance propagation robust to line renumbering.

Budget-Aware Core Packing. The packing of E_{raw} into E_{pack} is governed by a secondary knapsack reservation

$$\text{cost}(E_{\text{pack}}) \leq \beta_{\text{core}} \cdot B, \quad \beta_{\text{core}} \in [0.5, 0.9] \quad (9)$$

which ensures the greedy expansion phase retains working budget for relevance-propagated context. If $\text{cost}(E_{\text{raw}})$ ’s chosen representatives exceed $\beta_{\text{core}} \cdot B$, lowest-priority core fragments are downshifted to a more compact representative within their equivalence class P_u (e.g., signature-only variant for code, heading-only for prose)—a legal swap inside the matroid—prioritized by relevance score. A rescue sweep then retries still-skipped cores (cheapest first, with signature fallback) against the full remaining budget, and any core that still does not fit is demoted to an ordinary greedy candidate. Without this constraint, large diffs consume the entire budget on core fragments and the optimization pipeline degenerates.

4 Methodology

4.1 Instantiation: Typed Dependency Graph for Source Code

We now instantiate the framework of Section 3 for source code. Relevance weights $w(f, \Delta)$ are derived from a typed weighted graph $G = (\mathcal{V}, \mathcal{E}, w, \kappa)$ where nodes $\mathcal{V}$ correspond to fragments in F , edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ encode directed dependencies, $\kappa : \mathcal{E} \rightarrow T$ assigns each edge a type from a fixed set T (we reserve τ for the stopping threshold of Section 4.6), and $w : \mathcal{E} \rightarrow \mathbb{R}_{\geq 0}$ assigns a per-type weight (Section 4.3). Relevance is propagated by Personalized PageRank seeded at the diff (Section 4.2.1).

Pipeline phases. The implementation realizes Section 3’s formalism in two preliminary phases before any scoring runs. *File discovery* (`candidate_files.rs`) walks the working tree under `.gitignore`-aware filters and assembles the candidate file set used as the source of $\mathcal{V}$; this resolves the implicit “which files do we even consider” question that the formalism leaves to the implementation. *Fragment extraction* (`parsers/`, `fragmentation.rs`) then tree-sitter-parses each file into the fragments of Table 2, producing the fragment universe F . These two phases are distinct: file discovery operates on paths (no AST work), fragment extraction operates on file contents (no I/O). A third phase, *semantic discovery* (`discovery.rs`), is part of universe construction, not merely a scoring seed: the fragment universe is $F = \text{fragments}(\text{changed}) \cup \text{fragments}(\text{discovered})$, so a file the discovery ensemble does not return is never parsed, never enters the graph, and is unselectable at any budget. The deployed discovery is an ensemble (`DiscoveryKind::Ensemble`, `mode.rs`): related-file heuristics, rare-diff-identifier expansion, test-file discovery, and a BM25 lexical retriever contributing its top-1 file per query under graph modes (top-0 under internal-BM25 scoring mode, whose relevance signal already is lexical). Identifiers extracted from the commit message additionally seed the expansion-concept set (disabled via `DIFFCTX_NO_COMMIT_SIGNAL=1`); in the benchmark harness this signal is inert because gold patches are applied under a constant commit message, but in production use it is active. Two evaluation consequences follow: measured recall is upper-bounded by discovery coverage (quantified in Section 6.7), and the scoring-mode comparison of Section 6.5 holds the discovery ensemble fixed up to the top-1 BM25 contribution.

Fragment Types. Tree-sitter extraction yields 21 fragment kinds (Table 2), grouped by role: *full semantic units* (function bodies, class/struct/interface/enum/impl/type/module/variable/property/record definitions and declarations), *signature-only abridgements* of those (used as compact representatives in the partition matroid of Section 3), and *generic fallback* variants (`Section`, `Chunk`) for non-code text or files where parsing fails. Methods are encoded as `Function` fragments with the enclosing class providing context; only their signature variant (`MethodSignature`) is distinguished. Full units and their signature counterparts form the equivalence classes P_u of the matroid: at most one resolution per unit is selected.

Edge Categories. The set T comprises ten categories (Table 3): three core static-dependency categories (`Semantic`, `Structural`, `Similarity`); a `Sibling` category for file co-location (same-directory coupling); four extension categories that go beyond pure source-code dependencies (`Config`, `ConfigGeneric`, `Document`, `TestEdge`); the empirical `History` category mining co-change patterns from the git log [11]; and a `Generic` fallback for unclassified relations. `Config` subsumes domain-specific infrastructure cross-references (Helm charts, Kubernetes manifests, Dockerfile/Compose dependencies, CI/CD pipeline links, build-system configs); these share the same category multiplier per the per-category parameterization (Section 4.3). `Semantic` subsumes call/import/type/symbol-reference edges, which are individually distinguished (and individually weighted) at extraction time but share a single category multiplier. `TestEdge` links

Group	Variants
Full semantic units	Function, Class, Struct, Interface, Enum, Impl, Type, Module, Variable, Property, Record, Declaration, Definition
Signature-only	FunctionSignature, MethodSignature, ClassSignature, StructSignature, InterfaceSignature, EnumSignature
Generic fallback	Section, Chunk

Table 2: Fragment kinds extracted by the implementation. Full units and their signature counterparts form partition-matroid equivalence classes; generic fallback variants are used when tree-sitter parsing is unavailable or for non-code text.

tests to the code they exercise via static naming and import relations; the system does not execute tests, so runtime pass/fail signal is unavailable and weighting test edges by observed failures is future work. Three categories—**Semantic**, **Structural**, and **TestEdge**—are exempt from hub-suppression on the basis that their edges encode load-bearing structural and behavioral relationships that should not be dampened by node degree.

Category	Description
Semantic	Symbol references: function calls, identifier uses, type references, imports, Terraform module references.
Structural	AST containment (parent-child scope nesting).
Sibling	Files in the same directory (weak co-location coupling).
Similarity	Lexical/token overlap (TF-IDF on identifier tokens).
Config	Domain-specific infrastructure cross-references: Helm charts, Kubernetes manifests, Dockerfile/Compose dependencies, CI/CD pipelines, build-system configs (Bazel, CMake, etc.).
ConfigGeneric	Generic YAML/JSON cross-references not covered by domain-specific extractors.
Document	Cross-references in documentation (markdown links, reference-style citations).
History	Co-change patterns mined from git history.
TestEdge	Test-to-code relationships.
Generic	Fallback for unclassified edges.

Table 3: Edge categories in the typed dependency graph. Per-category multipliers w_t default to 1.0 pending calibration (Section 4.3); the structural-over-lexical prior lives in fine-grained per-edge-type weights.

Symbol Resolution Caveat. **Semantic** edges (call/import/type/symbol references) use AST queries and intra-scope lexical name matching as a heuristic approximation to data-flow analysis. True def-use chains require control-flow graph construction and reaching-definitions analysis, infeasible for incomplete or unparsable code in git working trees. Static resolution coverage is codebase-dependent and degrades with metaprogramming and dynamic dispatch (more severely in dynamic languages, less so in statically typed code with explicit annotations); quantifying resolution recall against runtime call graphs is left to empirical validation.

4.2 Relevance Scoring: A Pluggable Component

The framework of Section 3 is agnostic to the choice of $\hat{w}(f, \Delta)$. We instantiate it via three families spanning the design space—bounded local expansion (deployed default), global graph propagation, and pure lexical retrieval. All three are first-class scoring modes in the implementation; fine-grained per-edge-type weights are held fixed at literature-derived priors across modes (Section 4.3). Empirical comparison is reported in Section 5.

4.2.1 Personalized PageRank

We compute relevance scores via random walk with restart from core edits E_0 . Let s denote the restart distribution: $s(v) = 1/|E_0|$ for $v \in E_0$ and $s(v) = 0$ otherwise; s is a probability distribution by construction. The PPR fixed point is then

$$R_{\text{PPR}}(v) = (1 - \alpha) \cdot s(v) + \alpha \cdot \sum_{(u \rightarrow v) \in \mathcal{E}} R_{\text{PPR}}(u) \cdot \frac{w_{uv}}{\deg_{\text{out}}(u)} \quad (10)$$

where $\deg_{\text{out}}(u) = \sum_{u \rightarrow x} w_{ux}$ is the weighted out-degree, normalizing outgoing transition probabilities. Because dangling nodes drop their continuation mass (below), the deployed fixed point is *substochastic*—total mass is less than 1 before the final renormalization—so Eq. (10) should be read as the substochastic fixed point followed by a sum-to-one rescaling, applied *after* the forward–backward blending of Eq. (11).

The damping factor $\alpha \in [0.5, 0.65]$ controls locality. With continuation probability α , the number of *traversed edges* per walk before a restart follows a geometric distribution on $\{0, 1, 2, \dots\}$ with mean $\alpha/(1 - \alpha)$; equivalently, the number of *visited nodes* (including the seed) is $1/(1 - \alpha)$.¹ For $\alpha = 0.6$ this gives 1.5 traversed edges (or 2.5 visited nodes) before restart, sharper locality than web search ($\alpha = 0.85$, ≈ 5.7 edges), appropriate for the tighter dependency structure of code graphs. The implementation computes the fixed point with a forward-push (local-push) solver in the style of Andersen–Chung–Lang [1, 20] rather than global power iteration: per-node residuals are pushed until every residual falls below 10^{-4} . This per-coordinate criterion is a *termination tolerance*, not a direct score-error bound: the standard push invariant bounds the pointwise estimation error by the ℓ_1 norm of the residual vector, which the per-coordinate threshold controls only up to a factor of the touched-node count. A global push cap ($100 \cdot n$, bounded at 2×10^6) guards worst-case cost; runs that hit the cap are flagged as truncated and their relevance scores are biased toward the seeds. Dangling nodes (zero weighted out-degree) retain their restart mass but drop their continuation mass rather than teleporting it; the resulting deficit is absorbed by a final sum-to-one renormalization of the score vector.

Locality interpretation: In sparse code graphs with moderate branching, relevance decays approximately with graph distance, but the relationship is not strictly exponential on weighted directed graphs. Hub nodes with high in-degree can accumulate disproportionate mass regardless of semantic relevance, motivating the hub-suppression mechanism of Section 4.1.

Forward–Backward Blending. For diff understanding, both *forward* (callee-direction) and *backward* (caller-direction) propagation carry signal: forward PPR exposes the dependencies a changed function relies on, backward PPR exposes the call sites and tests impacted by it. The implementation runs PPR independently on the original graph and its transpose, then linearly blends the two scores:

$$R_{\text{PPR}}(v) = \rho \cdot R_{\text{PPR}}^{\text{fwd}}(v) + (1 - \rho) \cdot R_{\text{PPR}}^{\text{bwd}}(v), \quad \rho \in [0, 1] \quad (11)$$

¹Both quantities are reported in the literature; we adopt the edge-based convention throughout for consistency with the transition-matrix formulation.

where ρ is the forward-blend weight (default $\rho = 0.4$, i.e. backward-leaning, reflecting that for diff comprehension the impact-direction signal is typically stronger than the dependency-direction signal). The two PPR runs share the same α and convergence tolerance; only the edge orientation differs.

4.2.2 Bounded Ego-Network Expansion

Ego-network expansion computes relevance via depth-bounded BFS from the core edit set E_0 , with weights decaying by hop distance:

$$R_{\text{ego}}(v) = \sum_{u \in E_0} \mathbb{1}[d_{\text{hop}}(u, v) \leq L] \cdot \gamma^{d_{\text{hop}}(u, v)} \cdot W_{\text{path}}(u, v) \quad (12)$$

where $L \in \{1, 2\}$ is the expansion radius (`EgoGraphScoring.max_depth` in the implementation), $\gamma \in (0, 1]$ is a per-hop decay factor (`EgoScoringConfig.per_hop_decay`), $d_{\text{hop}}(u, v)$ is the number of edges in the shortest (by hop count) path from u to v in the edge-orientation-agnostic graph (BFS over the union of forward and reverse edges), and

$$W_{\text{path}}(u, v) = \max_{\pi \in \Pi_{=d_{\text{hop}}(u, v)}(u, v)} \prod_{(a, b) \in \pi} w_{ab} \quad (13)$$

is the maximum, over paths of shortest hop length, of the product of edge weights along the path. When the BFS traverses an edge against its stored orientation, the same stored weight is used—the reverse adjacency structure is an exact transpose with identical weights, and no traversal-time damping is applied; direction asymmetry arises only where an edge builder emits a separately weighted reverse edge (e.g. lexical-similarity edges carry a $0.5\times$ backward factor at construction time). We use product aggregation along paths to model multiplicative attenuation: relevance through a chain of weak edges decays compoundingly, mirroring PPR’s transition-probability semantics. Sum aggregation is an alternative; we report the product variant in evaluation.

In addition to the graph term, the implementation adds a small bounded lexical tie-break: non-core fragments sharing identifiers with the diff receive a bonus of up to $\varepsilon = 0.1$ (`DIFFCTX_EGO_LEXICAL_EPS`; overlap capped at 10 identifiers). EGO scoring is therefore predominantly structural but not purely graph-only, and the EGO-vs-BM25 comparison in Section 6 should be read accordingly; setting $\varepsilon = 0$ yields a graph-only ablation configuration.

Unlike PPR, ego-expansion has no global stationary distribution: relevance is strictly local. This differs from PPR in two computational respects: (i) cost is $O(|E_0| \cdot \bar{d}^L)$ where $\bar{d}$ is average degree, typically faster than PPR’s iterative convergence; (ii) hub nodes do not accumulate mass through long random walks, eliminating part of the motivation for hub-suppression on the scoring side.

Hypothesis on locality. Files relevant to a code change tend to lie within a small graph radius of the seed edit; relevance signal beyond depth 2 is dominated by noise from utility hubs and broad-capture configuration files. PPR with a high restart probability $1 - \alpha \in [0.35, 0.5]$ (continuation $\alpha \in [0.5, 0.65]$) effectively approximates EGO behavior, suggesting that for diff-aware context selection PPR’s global formulation may be over-engineered relative to the structural reality. Section 6 tests this empirically.

4.2.3 Lexical Baseline (BM25)

BM25 scoring treats fragments as documents and the diff as a query, ranking purely on lexical overlap of identifier tokens:

$$R_{\text{BM25}}(f) = \sum_{t \in q(\Delta)} \text{IDF}(t) \cdot \frac{\text{tf}(t, f) \cdot (k_1 + 1)}{\text{tf}(t, f) + k_1(1 - b + b \cdot |f|/\overline{|f|})} \quad (14)$$

where $\overline{|f|}$ is the mean fragment length over the candidate corpus, with parameters $k_1 = 2.5$, $b = 0.75$. The saturation parameter sits above the classical web-text range $k_1 \in [1.2, 2.0]$: identifier tokenization yields short documents with heavy term repetition, favoring later term-frequency saturation; a sensitivity sweep over k_1 is recorded as an open ablation. The query $q(\Delta)$ is the set of identifiers extracted from Δ . BM25 is the natural baseline: if structural reasoning provides no benefit, lexical retrieval should match graph-based methods. Reporting BM25 alongside graph methods is a methodological honesty check—without it, “graph-based context selection” is an unfalsifiable claim.

4.2.4 Discussion: What Each Mode Tests

The three scoring instantiations answer different questions in ablation:

- **BM25** establishes a lexical baseline and tests whether structural reasoning is necessary at all.
- **EGO** isolates the contribution of bounded local graph structure—“does the graph help, even just at depth 2?” This is the deployed default.
- **PPR** tests whether global relevance propagation provides additional signal beyond local expansion.

A monotone ordering $\text{BM25} < \text{EGO} < \text{PPR}$ would validate the structural-relevance hypothesis with maximum strength; non-monotone results (e.g., $\text{EGO} \approx \text{PPR}$ or $\text{EGO} > \text{PPR}$) inform task-structure conclusions about the radius of relevance in code change. Section 6 reports the comparison; the empirical result is non-monotone: internal BM25 leads headline recall (pooled 0.942) and nontrivial retrieval (0.402), PPR leads precision (0.328), EGO leads nontrivial retrieval among the graph-based modes (0.352 vs. 0.244), and the preregistered paired test finds no pairwise separation on headline recall (Section 6.5) — the relevance-signal choice trades off along different axes rather than dominating.

4.3 Edge-Type Weight Priors and Operational Calibration

We separate *structural* edge-type weights from *operational* hyperparameters of the selection algorithm.

Edge-type weights as fixed priors. Edge weights are parameterized in two layers. Fine-grained per-edge-type weights (e.g. call 0.55–0.70, inheritance 0.70–0.80, import 0.50–0.75, test-link 0.30–0.60, documentation 0.30; `config/weights.rs`) are author-selected priors informed by the code-navigation literature [15, 27]—those studies establish the structural-over-lexical *ordering*, not the specific coefficients—and are held fixed across modes and benchmarks. A second layer of per-category multipliers $w_t \in \mathbb{R}_{\geq 0}^{|T|}$ (one per category $t \in T$ of Table 3) currently defaults to 1.0 for every category, pending the calibration below. Full data-driven calibration over the $|T|$ -dimensional weight simplex,

$$w_t^* = \arg \max_{w_t \geq 0} \frac{1}{N} \sum_{i=1}^N \sum_{f \in C_i^*(w_t)} \mathbb{1}[f \in G_i] \cdot \omega_f, \quad (15)$$

is left to future work: the labeled corpora required to disambiguate ten weights without overfitting exceed the size used in the present evaluation. The above formulation is recorded here as the natural calibration target once such corpora become available; richer parameterizations such as learned per-edge weights $w(e) = \text{MLP}(\text{features}(e))$ over edge-context features remain a further direction.

Operational hyperparameters that are calibrated. The two scalar operational parameters of the selection algorithm are calibrated empirically:

- τ : the marginal-utility stopping threshold of the lazy-greedy loop (Section 4.6),
- β_{core} (a.k.a. `core_budget_fraction`): the fraction of the budget reserved for fragments adjacent to the core edit set E_0 before peripheral expansion.

These two parameters govern algorithm behavior at query time and admit a small grid search on a held-out manifest; calibration details are reported in Section 6. In the present work each scoring mode uses the same $(\tau, \beta_{\text{core}})$ values selected on the deployed-default (EGO) calibration; per-mode re-tuning is recorded as a sensitivity question in the limitations.

Per-source out-edge cap. The typed dependency graph admits power-law density on large repositories (django, mui/material-ui, microsoft/vscode), with raw edge counts reaching 3–5 M before any pruning. We cap each source node’s outgoing edge list to its top- K neighbors by post-suppression weight, applied *after* hub-suppression so the IDF damping computes against the full in-degree distribution. The default $K=64$ is the deployed value of the implementation; whether K admits a meaningful tuning gradient or can be treated as a structural constant is empirically open and we report a stratified analysis with paired statistical test in the appendix once a sufficiently powered comparison ($n \geq 500$ with Wilcoxon signed-rank) is completed. The implementation exposes the parameter via the `DIFFCTX_MAX_EDGES_PER_NODE` environment variable.

4.4 Temperature Calibration

The Boltzmann reformulation reduces the matroid+knapsack problem to a family of matroid problems parameterized by β : $\beta \rightarrow 0$ recovers cost-blind relevance maximization, $\beta \rightarrow \infty$ collapses to cost minimization. We propose a meta-level instance-conditional predictor:

$$\beta = f_{\theta}(B, \phi(\Delta, G_{\text{graph}})) \quad (16)$$

where features ϕ include diff token cost, fragment count, candidate-set size, graph density, and core-edit cost share. The predictor f_{θ} would be trained to minimize $|\text{cost}(C^*(\beta)) - B|$ on a validation corpus, with auxiliary recall floor against golden sets.

In the present work we describe the mechanism and implement deterministic calibration; we do not learn f_{θ} . The implementation selects β per-instance via geometric bisection over $[\beta_{\text{lo}}, \beta_{\text{hi}}]$ with the one-sided acceptance criterion $\text{cost}(C^*(\beta)) \geq B - \varepsilon$ (the symmetric $|\text{cost} - B| < \varepsilon$ collapses to this form because the budget cap saturates the upper side), which requires no training data. Bisection presupposes that $\text{cost}(C^*(\beta))$ is monotone non-increasing in β ; for the ranked-truncation selection this holds empirically but is not proven—re-ranking at nearby β can locally reorder the truncation boundary—so the loop maintains a one-sided bracket and is best read as heuristic root-bracketing rather than exact bisection on a certified monotone function. Replacement of binary search with a learned regression model is left to follow-up evaluation. Temperature calibration is a Lagrangian relaxation that affects budget targeting accuracy; it does not by itself recover any approximation guarantee on the original budgeted problem, and we do not claim such a recovery. The matroid-only $(1 - 1/e)$ row in Table 1 refers to continuous greedy on the matroid-only relaxation, not to the Boltzmann reweighting deployed here.

4.5 Submodular Extension via Concept Coverage

The modular weighted-recall objective generalizes to a submodular concept-coverage formulation, and this formulation is the *deployed default*: information needs Z (definitions, call targets, type signatures, impacted tests, background) are extracted from the diff by semantic analysis

(`needs_from_diff` in the implementation), with no human annotation required. Trading modularity buys the ability to model *diminishing returns*: once a concept is well-covered by one fragment, additional fragments documenting the same concept add less value. The modular form is recovered as the special case $Z = G$, $\phi = \text{id}$, $\text{rel}(f, z) = \mathbb{1}[f = z] \cdot \omega_z$, which collapses the coverage sum to weighted recall against golden fragments.

The extension is defined by:

$$U(C \mid \Delta) = \sum_{z \in Z} \phi \left(\max_{f \in C} \text{rel}(f, z) \right) \quad (17)$$

where $\text{rel}(f, z) = R(f)$ if fragment f contains/defines concept z , and $\phi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a nondecreasing function (e.g., $\phi(x) = \sqrt{x}$ or $\phi(x) = \min(x, 1)$). Concave ϕ enforces saturation; identity ϕ recovers the modular case.

Theorem 1. *Let $a_{f,z} \geq 0$ for all $f \in F, z \in Z$, and let $\phi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be nondecreasing with $\phi(0) = 0$, under the convention $\max_{f \in \emptyset} a_{f,z} = 0$ (so $U(\emptyset) = 0$). Then $U(C) = \sum_{z \in Z} \phi(\max_{f \in C} a_{f,z})$ is monotone submodular.*

Proof. Fix concept z and define $g_z(C) = \phi(\max_{f \in C} a_{f,z})$. For $C \subseteq T$, let $m_C = \max_{f \in C} a_{f,z}$ and $m_T = \max_{f \in T} a_{f,z}$. By set inclusion, $m_C \leq m_T$.

The marginal gain from adding fragment v to C is:

$$\Delta_C(v) = \phi(\max(m_C, a_{v,z})) - \phi(m_C)$$

Case 1: If $a_{v,z} \leq m_C$, then $\Delta_C(v) = 0$.

Case 2: If $a_{v,z} > m_C$, then $\Delta_C(v) = \phi(a_{v,z}) - \phi(m_C)$.

For $T \supseteq C$: If $a_{v,z} \leq m_T$, then $\Delta_T(v) = 0 \leq \Delta_C(v)$. If $a_{v,z} > m_T$, then since $m_T \geq m_C$ and ϕ is nondecreasing:

$$\Delta_T(v) = \phi(a_{v,z}) - \phi(m_T) \leq \phi(a_{v,z}) - \phi(m_C) = \Delta_C(v)$$

Thus g_z is submodular. Since $U = \sum_z g_z$ and each g_z is submodular, U is submodular (nonnegative linear combinations of submodular functions are submodular; nonnegativity of the g_z values themselves is not what is used).

Monotonicity: Adding fragments can only increase $\max_{f \in C} a_{f,z}$, and ϕ nondecreasing implies $U(C \cup \{v\}) \geq U(C)$. $\square$

The submodular objective is compatible with the partition matroid but breaks the Boltzmann reweighting machinery, which assumes modularity for the post-hoc truncation argument. In submodular mode, optimization uses direct knapsack with modified-greedy (Section 4.6, Option B), yielding $\frac{1}{2}(1 - 1/e)$ rather than $(1 - 1/e)$. This concept-coverage objective with $\phi(x) = \sqrt{x}$ is the deployed default; the modular Boltzmann-ranked variant is opt-in via an environment flag (`DIFFCTX_OBJECTIVE=boltzmann`). The objective-level ablation—submodular coverage vs. modular relevance at matched budgets—is reported in Table 9: the Boltzmann variant matches headline recall (0.928 vs. 0.919 pooled) but collapses precision to 0.084, confirming the concept-coverage objective as the deployed default.

File-Importance Prior for Impact Needs. For impact-type needs we apply a per-fragment importance prior $I(f) \in (0, 1]$ to downweight peripheral code: $m'(f, n) = m(f, n) \cdot I(f)$ when $n.\text{type} = \text{impact}$. Three categories are distinguished by directory pattern: generated code ($I = 0.10$), peripheral directories such as `examples/`, `demo/`, `vendor/`, `fixtures/`, `docs/` ($I = 0.15$), and default code ($I = 1.0$). Values reflect domain priors—generated code is mechanically derived from a schema and carries no human-intent signal; peripheral directories are typically out of scope for production review—and are not calibrated to the benchmark; a

$\pm 25\%$ / $\pm 50\%$ perturbation sensitivity check is recorded as an open ablation. Submodularity is preserved since $I(f) \in (0, 1]$ is a per-fragment static scalar applied before the max aggregation in $\phi(\max_f \text{rel}(f, z))$. Other need types (definition, signature, test, invariant, background) use unmodified $m(f, n)$ since they admit no analogous production-vs-peripheral distinction.

4.6 Greedy Selection with Adaptive Stopping

We employ a density-based greedy algorithm with lazy evaluation [21] for computational efficiency.

Two regimes. The greedy procedure operates in two regimes corresponding to how the budget is enforced.

Regime A (cost-penalized surrogate ranking). Score fragments by the cost-penalized $\tilde{w}(e) = w(e) \exp(-\beta|e|)$ and rank within each partition class; the budget is enforced post-hoc by truncation. This is a Lagrangian-style ranking heuristic, not an approximation algorithm for the original budget-constrained objective; we report it as one operating mode.

Regime B (explicit knapsack, current implementation). Density-greedy on the concept-coverage $\mathcal{U}$ of Section 4.5 with $\text{cost}(C) \leq B$, plus two engineering bonuses: a modular structural bonus ($0.10 \cdot r_{\text{norm}}$ for fragments matching at least one need), which preserves submodularity, and a coverage-saturation (“diversity”) bonus whose per-fragment increment is pointwise non-increasing in the selected set but path-dependent, falling outside the strict set-function form (Heuristics Annex); setting `DIFFCTX_RELATEDNESS_BONUS=0` disables the path-dependent term for clean analyzable runs. Without the adaptive-stopping heuristic and bonuses, the modified-greedy variant (Option B below) admits $\frac{1}{2}(1 - 1/e)$ on the monotone submodular-knapsack instance [14]. With adaptive stopping (deployed default), the variant is heuristic; Proposition 1 bounds the utility foregone by stopping.

Regime B is the default path. Regime A is opt-in via the `DIFFCTX_OBJECTIVE=boltzmann` environment flag, applying β -reweighting with binary-search calibration over the modular relevance scores, used as a budget-aware ranking signal rather than an approximation guarantee. Comparison between the two regimes is empirical (Section 5).

Approximation guarantees. For monotone submodular maximization, the classical greedy algorithm achieves a $(1 - 1/e) \approx 0.632$ approximation under **cardinality constraints** ($|C| \leq k$) [23]. Our problem uses a **knapsack constraint** ($\text{cost}(C) \leq B$ with heterogeneous fragment costs), for which density-greedy does not admit the same guarantee. We therefore present two options:

Option A (Heuristic): Use density-greedy as a practical heuristic without worst-case bounds. We hypothesize that performance on real code graphs may exceed 0.5 of optimal due to favorable structure, but this requires empirical validation.

Option B (Modified Greedy with Guarantee): Return the better of (i) the density-greedy solution C_{greedy} and (ii) the best single addition to the packed core in the E_{pack} -contracted instance:

$$C_{\text{out}} = \arg \max \{ U(C_{\text{greedy}}), \max_{f: E_{\text{pack}} \cup \{f\} \in \mathcal{F}} U(E_{\text{pack}} \cup \{f\}) \}$$

Because the deployed problem is contracted at E_{pack} (Section 3), the singleton comparison must be made in the contracted instance— $E_{\text{pack}} \cup \{f\}$ under the residual budget and the partition/overlap feasibility—not against a bare singleton in the original universe. This modification provides the constant-factor guarantee on the *knapsack-only* contracted instance [14, 16].

Note: the guarantee applies only to the variant *without* adaptive stopping, and it does not transfer to the deployed matroid $\cap$ knapsack feasible region (Table 1 lists matroid $\cap$ knapsack guarantees only for continuous-greedy variants we do not deploy); the stopping heuristic is evaluated empirically.

Algorithm 1 Lazy Greedy Context Selection (Heuristic Version)

```
1: Pack  $E_0$  (relevance-descending) under  $\beta_{\text{core}} \cdot B$ , downshifting oversized cores to signature
   variants; rescue-sweep skipped cores against the full budget;  $C \leftarrow$  packed cores
2:  $Q \leftarrow$  priority queue of all remaining candidates by marginal density
3:  $d_{\text{peak}} \leftarrow 0$ 
4: while  $\text{cost}(C) < B$  and  $Q \neq \emptyset$  do
5:    $f^* \leftarrow$  pop highest-density candidate from  $Q$ 
6:   if  $f^*$  evaluation is stale then
7:     Recompute density, reinsert into  $Q$ 
8:     continue
9:   end if
10:  if  $\text{density}(f^*) > d_{\text{peak}}$  then
11:     $d_{\text{peak}} \leftarrow \text{density}(f^*)$ 
12:  else if  $d_{\text{peak}} > 0$  and  $\text{density}(f^*) < \tau \cdot d_{\text{peak}}$  then
13:    break ▷ Adaptive stopping (heuristic; Prop. 1)
14:  end if
15:  if  $\text{cost}(C) + |f^*| \leq B$  then
16:     $C \leftarrow C \cup \{f^*\}$ 
17:  end if
18: end while
19: return better of  $C$  and the best core-contracted single addition (Option B)
```

Adaptive stopping (engineering heuristic). The stopping threshold $\tau \in [0.04, 0.16]$ (the grid-explored range; `DEFAULT_STOPPING_THRESHOLD` in `config/limits.rs`, overridable via the `--tau` CLI flag; deployed calibrated default 0.12) terminates expansion when marginal utility-per-cost falls below a fraction of the running peak density d_{peak} . This modification improves token efficiency in practice but forfeits the constant-factor guarantee against the offline optimum, as greedy analysis assumes selection continues to the feasibility frontier. It does, however, admit an additive certificate:

Proposition 1 (Additive stopping certificate, greedy phase). *Run density-greedy over a candidate queue Q on a monotone objective whose per-candidate marginal gain is non-increasing in the selected set—true for the monotone submodular $\mathcal{U}$ of Section 4.5, and preserved by the deployed bonuses (the structural bonus is modular; the diversity bonus’s increment is pointwise non-increasing in C). Let d_{peak} be the maximum marginal density observed during the greedy expansion phase and B_{rem} the unspent budget at the stopping point. If the loop stops when the best available candidate has density below $\tau \cdot d_{\text{peak}}$, then continuing the same greedy over Q to the feasibility frontier would increase the objective by less than $\tau \cdot d_{\text{peak}} \cdot B_{\text{rem}}$.*

Proof. At the stopping point every candidate in Q has current density at most that of the popped best, hence below $\tau \cdot d_{\text{peak}}$. Marginal gains are non-increasing in the selected set, so any fragment f the continued greedy would accept is accepted at density below $\tau \cdot d_{\text{peak}}$, contributing marginal utility below $\tau \cdot d_{\text{peak}} \cdot |f|$. The accepted fragments’ total cost is at most B_{rem} ; summing yields the bound. $\square$

Scope of the certificate. The certificate applies to the *intermediate* greedy-phase output, before the post-passes of the Heuristics Annex, and is relative to the greedy continuation, not to the offline optimum (the constant-factor discussion of Table 1 concerns the latter and applies to the variant without stopping). Its reach over post-pass additions is determined by where their candidates come from. The *coherence pass* draws only from the post-filter candidate universe that populated Q ; at the stopping point every such fragment has density below $\tau \cdot d_{\text{peak}}$, so

coherence additions within the remaining budget stay inside the same $\tau \cdot d_{\text{peak}} \cdot B_{\text{rem}}$ bound. The *rescue pass* and the *changed-files pass*, by contrast, draw from the pre-filter fragment universe—fragments that were never scored into Q and carry no density bound—and the changed-files pass adds without a relevance gate (within budget; a changed file whose representatives do not fit stays unrepresented and surfaces as retention < 1). The certificate does not extend to those additions; they are monotone utility gains, but unquantified ones. Note the coherence coverage is airtight with respect to size: the budget remainder only decreases over the run, so a candidate once discarded for size can never fit later, and any coherence addition that fits the remaining budget was necessarily still in Q at the stopping point, inheriting the density bound. The implementation emits the bound per run (`stopping_certificate` = $\tau \cdot d_{\text{peak}} \cdot B_{\text{rem}}$ in the analytics output) whenever the loop terminates by threshold. The with-vs-without-stopping ablation at matched budgets is reported in Table 9: disabling stopping ($\tau=0$, there is no dedicated flag) moves pooled recall and precision by under one point at $B=8000$, so adaptive stopping earns its keep on latency rather than quality. The deployed calibrated default is $\tau = 0.12$ (Section 6); $\tau \approx 0.08$ for narrowly-scoped review remains an uncalibrated heuristic suggestion.

Complexity. The algorithm runs in $O(|\mathcal{V}| \log |\mathcal{V}| + k \cdot T_{\text{eval}})$ where $k = |C|$ (typically 15–50 fragments) and $T_{\text{eval}} = O(|Z|)$. Lazy evaluation significantly reduces recomputations versus standard greedy on sparse code graphs.

5 Evaluation Methodology

Since “understanding” is latent and task-dependent, we operationalize it via a behavioral proxy with directly measurable outcomes: *file-level recall against an annotated golden context* on multi-language software-engineering benchmarks. Section 6 reports results under this protocol; additional proxies (dependency coverage, co-change recall, fault-localization accuracy) are described as natural extensions in Section 8.1.

5.1 Primary Metric: File Recall at Fixed Token Budget

For a diff Δ with golden file set G and selector output C (the selected fragments projected to their containing files) subject to $\text{cost}(C) \leq B$, define

$$\text{Recall}(C, G) = \frac{|C \cap G|}{|G|}, \quad \text{Precision}(C, G) = \frac{|C \cap G|}{|C|}. \quad (18)$$

At a fixed token budget B , recall is the load-bearing metric: precision is constrained by $|G|/k$ where k is the number of files that fit, and is reported only for completeness. The metric is independent of which scoring instantiation $\hat{w}$ is active and is computed identically across all three modes (EGO, PPR, BM25) and across baselines.

Operationalization of G per benchmark. The golden set is constructed differently per benchmark, and the difference determines what recall measures. Throughout, $\text{files}(\Delta)$ denotes the post-apply file paths touched by the gold patch (pure deletions and rename-source paths excluded), and fragment-level annotations are projected to their containing files. *SWE-bench Verified* ships no context annotation, so $G = \text{files}(\Delta)$ by construction. *PolyBench-500* ships CST-node annotations (`modified_nodes`) that all lie inside patch files, so its file-level projection again coincides with $\text{files}(\Delta)$. *ContextBench Verified* ships fragment-level golden annotations whose file projection genuinely extends beyond the patch: on the 500-instance test manifest, 218 instances (43.6%) have at least one gold file outside $\text{files}(\Delta)$, and 594 of 2,072 gold files (28.7%) lie outside the diff (213 instances / 573 files among *ok*-status rows at $B=8000$ — the subset the retrieval analysis of Section 6.7 reports). Consequently, file recall measures *seed retention* on SWE-bench Verified and PolyBench-500, and *dependency-context retrieval* only on ContextBench Verified. Two derived metrics follow: *seed-excluded recall* (`nontrivial_file_recall`), $|C \cap (G \setminus \text{files}(\Delta))|/|G \setminus \text{files}(\Delta)|$, defined only where the denominator is nonempty (in practice: ContextBench Verified), and *changed-file retention*, $|C \cap \text{files}(\Delta)|/|\text{files}(\Delta)|$, defined everywhere. Both are recorded per instance by the harness and reported in Section 6.7.

5.2 Test Sets and Benchmarks

Evaluation uses three frozen manifests aggregating instances from publicly released benchmarks: SWE-bench Verified [13, 24], a curated 500-instance subset of SWE-PolyBench [26], and the fragment-annotated ContextBench Verified configuration [18]. Stratification is performed per (`source_benchmark`, `language`); cross-benchmark contamination is filtered by (`repo`, `base_commit`) key. Manifests are committed to git before any calibration trial executes. HuggingFace dataset revisions are pinned by SHA so that subsequent reruns operate on the same data.

5.3 Calibration Protocol

Two operational parameters are calibrated on a separate calibration manifest disjoint from the test manifests: τ (stopping threshold) and β_{core} (core budget fraction), as defined in

Section 4.3. The calibration objective is the minimum file-recall across the four calibration source benchmarks,

$$\text{score}(\tau, \beta_{\text{core}}) = \min_{b \in \mathcal{B}_{\text{cal}}} \text{recall}_b(\tau, \beta_{\text{core}}), \quad (19)$$

chosen to avoid bias toward the largest-population calibration benchmark. A grid search of 5×3 values over $(\tau, \beta_{\text{core}})$ is then validated on a held-out 235-instance validation manifest before fixing parameters and running the test set exactly once.

5.4 Baselines

Two external baselines are implemented at the same fixed token budget B on identical manifests:

- **BM25 over patch identifiers** (`benchmarks/baselines/bm25_baseline.py`): rank-bm25 over all repo files, query = identifiers extracted from the gold patch, file-pack to budget. This isolates pure lexical retrieval without graph structure and corresponds to the SWE-bench paper’s reference protocol [13] adapted to a tight budget.
- **Aider repo-map** [9] (`benchmarks/baselines/aider_baseline.py`): the closest publicly available approach using the same primitives (tree-sitter AST extraction + PageRank + token-budgeted selection). To avoid an unfair information leak, we run Aider in two modes, both at the same $B=8000$ token budget on the same manifests:
 - *Fair-input Aider*: `mentioned_fnames` restricted to file paths that already appear in the input diff text (the same information `diffctx` receives from the patch). `mentioned_idents` extracted from the patch identically to how it is computed for the BM25 baseline.
 - *Oracle-mentioned Aider*: `mentioned_fnames` populated from `instance.gold_files`. This is **not** a baseline but an approximate upper-bound stress test — it is what Aider would produce if a perfect oracle named the gold files in advance.

The fair-input mode is the headline comparison; the oracle mode bounds how much downstream value file-hint information could provide to a PPR-on-tree-sitter approach.

Sourcegraph Cody [10] is cited as related work but cannot be directly compared: its evaluation suite is internal and requires the Sourcegraph backend.

6 Empirical Evaluation

This section instantiates the protocol of Section 5 and reports the consolidated rerun at git tag `paper-v2` (v4 calibration, status-gated patch application, two-pass edge construction). All cells have completed on the full 1500-instance test set: the three scoring modes on the full budget grid, four operational ablations, the two trivial floors, and the external baselines (whole-file BM25 on the budget grid, Aider fair-input, and the Aider oracle-mentioned stress test at $B=8000$).

6.1 Setup

Test sets. Evaluation uses three frozen manifests, 500 instances each (1500 total), stratified per (`source_benchmark`, `language`):

- *SWE-bench Verified* [13, 24] — 500 Python instances drawn from 12 unique repositories.
- *PolyBench-500* [26] — a 500-instance multi-language curated subset of SWE-PolyBench, spanning 21 unique repositories with 125 instances each in Python, Java, JavaScript, and TypeScript.
- *ContextBench Verified* [18] — 500 instances spanning 58 unique repositories with fragment-level golden annotations across 8 languages (Python 266, TypeScript 66, JavaScript 60, Go 40, C 23, Rust 20, Java 15, C++ 10; counts verified directly against the frozen manifest, matching the per-language totals of Table 6: $140 = 125 + 15$).

Cross-benchmark contamination is filtered by (`repo`, `base_commit`) key (1449 instances dropped from the calibration pool). HuggingFace dataset revisions are pinned by SHA in `benchmarks/dataset_revisions.json`.

Calibration. The v4 re-calibration on the deployed-default EGO pipeline runs a 5×3 grid over $\tau \in \{0.04, 0.08, 0.12, 0.16, 0.20\}$ and $\beta_{\text{core}} \in \{0.5, 0.7, 0.9\}$ on the v1 calibration manifest (600s per-instance timeout, per-cell logs retained as an artifact). The selection objective is the minimum file-recall across the four calibration source benchmarks (Section 5); minimum rather than mean prevents domination by the largest-population benchmark. The full grid is reported in Table 4. The surface is flat and cleanly monotone: the top-3 cells lie within 0.004, $\beta_{\text{core}}=0.5$ dominates every row, and the objective degrades monotonically in both parameters — $(\tau, \beta_{\text{core}})$ are robust rather than tuned to a sharp optimum. The grid-best cell is $\tau=0.04$ (0.6572); validation on the held-out manifest is statistically indistinguishable across the flat top and confirms $\tau=0.12$, $\beta_{\text{core}}=0.5$ as the deployed winner (validation score 0.6604), which all rerun cells use. An earlier 4×3 grid under the since-removed v1 hybrid combiner (winner score 0.1092 under a stricter objective normalization; git tag `paper-v1-grid`) showed the same flatness and the same winning cell, so the v4 grid is a confirmation, not a change of operating point. The remaining operational parameters (Section 4.6 and the appendix) are held fixed at the values listed there.

Reproducibility. Test, validation, and calibration manifests were frozen and committed to git (tag `paper-v2-freeze`, “*feat(benchmarks): freeze v1 splits*”, 2026-04-29) before any calibration trial executed. Calibration trials checkpoint to disk; final test runs are conducted exactly once on the frozen splits. Stratification seed is 42 with validation fraction 10%; details in `benchmarks/manifests/v1/SPLIT_REPORT.md`. Hardware: Apple M4 Pro, 48 GB unified memory, macOS arm64; Python 3.12.12 with dependencies pinned in `requirements-bench.lock` (`uv pip compile`); Rust toolchain pinned to 1.92.0 via `rust-toolchain.toml`; per-instance timeout 900s for sweep cells (600s for calibration, 600s for the Aider helper request); checkpoint/resume supported. Bootstrap CIs are stable under seed variation (seeds 42/43/44 agree

Table 4: Calibration grid scores (v4 re-calibration on the deployed-default EGO pipeline): minimum per-source-benchmark file recall on the v1 calibration manifest, $B=8000$, 5×3 grid over $\tau \in \{0.04, \dots, 0.20\}$ and $\beta_{\text{core}} \in \{0.5, 0.7, 0.9\}$. The validated winner ($\tau=0.12$, $\beta_{\text{core}}=0.5$) is highlighted. The surface is flat and cleanly monotone: the top-3 cells lie within 0.004, $\beta_{\text{core}}=0.5$ dominates every row, and the objective degrades monotonically in both parameters; the grid-best cell ($\tau=0.04$, 0.6572) is statistically indistinguishable from the winner on the held-out validation manifest, which confirmed $\tau=0.12$.

$\tau \setminus \beta_{\text{core}}$	0.5	0.7	0.9
0.04	0.6572	0.6518	0.6431
0.08	0.6554	0.6507	0.6420
0.12	0.6532*	0.6483	0.6402
0.16	0.6494	0.6442	0.6357
0.20	0.6459	0.6417	0.6327

to four decimals on the pooled headline; `artifacts/stats_robustness.json`). Committed alongside this paper under `docs/Context-Selection-for-Git-Diff/v2/artifacts/`: per-cell aggregate JSON, per-instance CSVs for every $B=8000$ cell including the external BM25 baseline (so the preregistered P1 test is reproducible from the repository), the fusion-union per-instance CSV with deduplicated token costs, and the cluster-bootstrap/multi-seed robustness statistics — every table value traces to a versioned artifact.

6.2 Headline Results: EGO Mode (Deployed Default)

Table 5 reports per-benchmark file-level recall and precision over the full sweep (all three test sets at 500/500). Table 6 breaks the same data down by source language with non-parametric percentile bootstrap 95% confidence intervals ($B=10,000$ resamples, seed=42).

Table 5: Per-benchmark file-level metrics, scoring=EGO (deployed default), $B=8000$ tokens, with 95% percentile bootstrap CIs ($B=10,000$ resamples, seed=42). Consolidated rerun at git tag `paper-v2` (v4-calibrated $\tau=0.12$, $\beta_{\text{core}}=0.5$, fixed patch-apply status gating, two-pass edge construction). Reported as *intent-to-treat*: recall and precision are computed over all $n=500$ instances per test set, with `clone_fail`, `apply_fail`, and `timeout` rows contributing recall 0. The `ok%` column reports the fraction of instances where the pipeline produced a non-error selection. An *ok-only* sensitivity analysis (recall over the `ok%` subset) gives 0.840 on ContextBench Verified, 0.988 on PolyBench-500, and 0.998 on SWE-bench Verified; the residual non-ok rows are 5 deterministic dataset-side patch-encoding failures (CRLF-in-repo vs. LF-normalized gold patch) plus per-instance timeouts in the near-dense-graph repository class. We keep ITT as the headline for honesty. The pooled row is the instance-level mean; it coincides with the macro (per-set) average because each test set has $n=500$. It is *not* the IR micro-average $\sum_i |C_i \cap G_i| / \sum_i |G_i|$, which weights instances by gold-set size and differs from both.

Test set	n	File recall	File precision	ok%
ContextBench Verified	500	0.816 [0.792, 0.839]	0.236 [0.216, 0.256]	97.2%
PolyBench-500	500	0.942 [0.923, 0.960]	0.248 [0.227, 0.271]	95.4%
SWE-bench Verified	500	0.998 [0.994, 1.000]	0.158 [0.144, 0.174]	100.0%
Pooled	1500	0.919 [0.908, 0.929]	0.214 [0.203, 0.226]	97.5%

Table 6: Per-language metrics with 95% percentile bootstrap CIs ($B=10,000$). Pooled across the three test sets, *intent-to-treat*: instances whose pipeline run failed (clone, apply, timeout) contribute recall 0. The ok-only column reports mean recall over the n_{ok} instances with a completed selection. The out-of-memory kill class that dominated Go in the previous snapshot (21/40 instances) is resolved by the two-pass bounded edge construction; the remaining Go gap to ok-only is per-instance timeouts on near-dense-graph repositories.

Language	n	n_{ok}	File recall (ITT)	File recall (ok-only)
Python	891	884	0.938 [0.926, 0.949]	0.945
TypeScript	191	176	0.879 [0.837, 0.919]	0.954
JavaScript	185	185	0.917 [0.889, 0.942]	0.917
Java	140	131	0.903 [0.856, 0.945]	0.965
Go	40	34	0.782 [0.664, 0.887]	0.921
C	23	23	0.885 [0.811, 0.948]	0.885
Rust	20	20	0.944 [0.885, 0.989]	0.944
C++	10	10	0.809 [0.701, 0.917]	0.809

6.3 Observations

Recall at a tight 8k-token budget. Pooled mean file recall is 0.919 [95% CI 0.908, 0.929] across all 1500 instances at $B=8000$. Per-benchmark recall is highest on SWE-bench Verified (0.998), then PolyBench-500 (0.942), and lowest on ContextBench Verified (0.816); the per-set spread is much larger than each set’s CI half-width and is therefore benchmark-driven, not noise. SWE-bench Verified file recall saturates near 1.0 across the entire budget grid {8k, 16k, 32k, 64k, 128k, -1 }; Section 6.7 shows that this saturation is a property of the gold-set composition (the gold set coincides with the input diff’s own files on that benchmark), so the headline recall on SWE-bench Verified and PolyBench-500 measures *changed-file retention* rather than context retrieval, and ContextBench Verified is the only test set whose recall reflects genuine retrieval.

Per-language pattern. On ok-only recall, Java (0.965), TypeScript (0.954), Python (0.945), and Rust (0.944) lead; Go (0.921) and JavaScript (0.917) follow closely; C (0.885) and C++ (0.809) trail with sample sizes too small for firm conclusions. The out-of-memory kill class that dominated Go in the previous snapshot (21/40 instances, `status=oom_kill`) is resolved: the two-pass edge construction with a bounded per-source top- K (output-equivalent by construction, verified by the bit-equivalence gate) cut retained edge state from $O(E_{\text{raw}})$ to $O(n + Kn)$, and 34 of 40 Go instances now complete; the residual gap is per-instance timeouts on near-dense-graph repositories. The per-language spread on completed instances is under 6 points among $n \geq 40$ languages—consistent with the symbol-reference heuristic handling the common cases despite its known blindness to dynamic dispatch (Section 4.2); a per-language seed-excluded breakdown remains future work before drawing a resolver-quality conclusion.

Precision remains low (≈ 0.21 pooled). Mean file precision is approximately $|G|/k$ where $|G|$ is the gold-file count per instance ($\sim 1\text{--}3$) and k is the number of files that fit the budget. Low file precision indicates that the selector retrieves many non-gold files under the fixed budget; downstream LLM context quality depends on both recall and noise, so file recall is a necessary but insufficient proxy. The scoring-mode ablation (Section 6.5) shows precision is strongly mode-dependent: PPR reaches 0.328 pooled at essentially identical recall, so the EGO precision

deficit is a property of the expansion policy, not of the budget. ContextBench Verified provides block-level (line-range) annotations beyond the file level; we plan to report block-level recall on that subset to characterize how concentrated the selected context is around the gold regions, and to bound the true noise floor implied by the file-precision number.

6.4 Baseline Comparisons

Two external file-level baselines run the same protocol (same manifests, same budget, same metric) so that the comparison is apples-to-apples; the Aider oracle-mentioned mode (an upper-bound stress test, not a baseline) is additionally reported in the caption of Table 7. Methodology follows the standard IR significance protocol [28]: per-instance paired bootstrap on the recall delta ($B=10,000$ resamples, percentile CI) and a one-sided permutation sign-flip test (`permutation_paired`, 10^5 permutations, add-one corrected — exact rather than the asymptotic Wilcoxon approximation whose nominal p -values fall meaninglessly below machine precision). Repository-level cluster inference matters because the 500 SWE-bench instances come from only 12 repositories and are not independent: re-estimating the P1 delta with the harness’s `paired_cluster_bootstrap` (clusters resampled by repository) widens the SWE-bench CI from $[+0.415, +0.499]$ to $[+0.384, +0.607]$ around the same $+0.457$ mean — the conclusion is unchanged under the conservative estimator, and instance-level CIs elsewhere in the paper should be read as anti-conservative on that test set.

BM25 over patch identifiers. `rank-bm25` indexes all repository files; the query is the set of identifiers extracted from the gold patch (camelCase + snake_case decomposition, language-keyword filter). Files are packed in BM25 rank order under strict greedy budget compliance. Implementation in `benchmarks/baselines/bm25_baseline.py`. This isolates pure lexical retrieval without any graph structure.

Aider repo-map. The reference open-source approach using the same primitives (tree-sitter AST + Personalized PageRank + token-budgeted packing). Invoked through an isolated `uv tool` subprocess with the upstream `aider-chat==0.86.2` package. We run Aider in two modes: *fair-input* (`mentioned_fnames` restricted to file paths visible in the input diff text, the same information `diffctx` receives) and *oracle-mentioned* (`mentioned_fnames` populated from `instance.gold_files`, an upper-bound stress test rather than a baseline). Implementation in `benchmarks/baselines/aider_baseline.py`.

Pairing. Both baselines run on the same 1500 instances. Method columns in Table 7 are intent-to-treat means (failed rows contribute recall 0); the paired Δ columns follow the preregistered P1 protocol and pair on the *both-OK* subset ($n_{\text{both-OK}} = 1460$ for BM25, 1421 for Aider fair-input), so an infrastructure failure on one side is never scored against the other side’s real recall. Failure rates are not symmetric — `diffctx` completes 97.5% pooled, Aider fair-input 96.6% pooled (worst on PolyBench-500 at 91.6%, where its bundled tree-sitter grammars mismatch several languages) — which is why the both-OK pairing matters.

6.5 Scoring-Mode Ablation

Section 4.2 introduces three pluggable scoring modes (EGO, PPR, internal BM25). In the consolidated rerun all three are run in isolation inside the pipeline on the full 1500-instance test manifest with operational hyperparameters held at the EGO-default values ($\tau=0.12$, $\beta_{\text{core}}=0.5$, $B=8000$). The internal-BM25 cell (`BM25Scoring` on the identical candidate universe, fragments, core handling, and post-passes) closes the gap flagged in the previous snapshot: Table 8 is now a genuine within-framework source-of-relevance ablation, holding the selection machinery constant and varying only $\hat{w}(f, \Delta)$. Per-mode re-tuning is recorded as a sensitivity question —

Table 7: diffctx (EGO, deployed default) vs. external baselines at $B=8000$ tokens. Method columns are intent-to-treat means over all $n=500$ per test set; Δ is the per-instance paired delta in file recall on the both-OK subset per the preregistered P1 protocol ($n_{\text{both-OK}}=1460$ pooled for BM25), positive favors diffctx. 95% paired-bootstrap percentile CI on Δ ($B=10,000$ resamples, seed = 42); p -value from the permutation sign-flip test (`permutation_paired`, one-sided, 10^5 permutations, add-one corrected; $1.0e-5$ is the add-one floor). Both columns from the consolidated rerun at git tag `paper-v2` ($n_{\text{both-OK}}=1421$ pooled for Aider fair-input, measured with the helper `stderr-deadlock` fix). The Aider oracle-mentioned stress test (`mentioned_fnames` populated from the gold files — an upper bound, not a baseline) reaches pooled ITT recall 0.505 [0.482, 0.528]; diffctx-EGO exceeds even this oracle upper bound by +0.410 [+0.385, +0.434] paired, resolving the preregistered P3 question decisively: repo-map packing loses gold files even when literally told which they are.

Test set	n	diffctx	baseline	Δ recall [95% CI]	perm. p
<i>vs. BM25 over patch identifiers</i>					
ContextBench Verified	500	0.816	0.485	+0.352 [+0.320, +0.386]	1.0e−5
PolyBench-500	500	0.942	0.667	+0.300 [+0.268, +0.332]	1.0e−5
SWE-bench Verified	500	0.998	0.540	+0.457 [+0.415, +0.499]	1.0e−5
Pooled	1500	0.919	0.564	+0.371 [+0.350, +0.392]	1.0e−5
<i>vs. Aider repo-map (fair-input mode)</i>					
ContextBench Verified	500	0.816	0.468	+0.353 [+0.315, +0.391]	1.0e−5
PolyBench-500	500	0.942	0.316	+0.635 [+0.595, +0.674]	1.0e−5
SWE-bench Verified	500	0.998	0.694	+0.304 [+0.266, +0.344]	1.0e−5
Pooled	1500	0.919	0.493	+0.423 [+0.399, +0.447]	1.0e−5

if the deployed-default hyperparameters disadvantage a single-mode run, that disadvantage is itself an argument against the single mode in deployment.

Section 4.2.2 predicts that PPR with a high restart probability $1 - \alpha$ approximates bounded-radius EGO. The ablation tests this empirically: a monotone ordering $\text{BM25} \prec \text{EGO} \prec \text{PPR}$ would validate the structural-relevance hypothesis with maximum strength; the observed outcome (below) is more nuanced and motivates the retrieval-vs-retention decomposition of Section 6.7.

The ablation reveals three structural facts. First, no single mode dominates: internal BM25 leads headline recall (0.942 pooled, at 40% fewer used tokens), PPR leads precision (0.328 pooled, +11 points over EGO at essentially identical recall), and the deployed default EGO leads neither. Second, on SWE-bench Verified all modes saturate at recall ≈ 1.0 ; as Section 6.7 shows, gold coincides with the input diff’s own files there, so the mode ordering on that set is uninformative about retrieval. Third, the headline ordering *inverts* on the retrieval-only metric: on the ContextBench Verified subset with gold files outside the diff, EGO recovers 0.352 of nontrivial gold versus 0.402 for internal BM25 and 0.244 for PPR, while the union of EGO and BM25 hits reaches 0.473 — the two signals are complementary rather than redundant (Section 6.7). The preregistered P2 test on the both-OK subset ($n=1462$) quantifies how little the *headline* metric separates the modes: the Friedman omnibus rejects exchangeability ($\chi^2 = 39.2$, $p = 3.1 \times 10^{-9} < 0.0167$ after Holm), yet no pairwise Nemenyi comparison clears the critical difference (CD = 0.087; largest rank gap 0.063), and the paired recall deltas are within ± 0.006 pooled (internal BM25 − EGO: +0.006 [+0.002, +0.009]; PPR − EGO: −0.006 [−0.010, −0.003]). Headline recall is retention-saturated; the operative differences between modes live in precision (Table 8) and nontrivial retrieval (Table 13).

The clean-run operational ablations promised by the methodology (Section 4.6) were also

Table 8: Scoring-mode ablation at $B=8000$, deployed-default operational hyperparameters, consolidated rerun. File recall (upper block) and file precision (lower block) with 95% percentile bootstrap CIs, intent-to-treat. All three columns are instantiations of $\hat{w}(f, \Delta)$ inside the identical diffctx pipeline (same fragmentation, core handling, and post-passes; candidate universes are identical up to the discovery ensemble’s top-1 BM25 contribution, which is enabled for the graph modes and disabled under internal-BM25 scoring — a difference of at most one file per query); EGO is the deployed default, and internal BM25 is `src/scoring.rs::BM25Scoring` — distinct from the external whole-file-packing baseline of Table 7. Mean used tokens at $B=8000$: EGO 4915, PPR 4904, internal BM25 2883.

Test set	n	EGO (default)	PPR	BM25 (internal)
<i>File recall</i>				
ContextBench Verified	500	0.816 [0.792, 0.839]	0.806 [0.782, 0.829]	0.846 [0.825, 0.866]
PolyBench-500	500	0.942 [0.923, 0.960]	0.956 [0.938, 0.971]	0.981 [0.970, 0.990]
SWE-bench Verified	500	0.998 [0.994, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
Pooled	1500	0.919 [0.908, 0.929]	0.920 [0.910, 0.930]	0.942 [0.934, 0.951]
<i>File precision</i>				
ContextBench Verified	500	0.236 [0.216, 0.256]	0.357 [0.331, 0.383]	0.284 [0.263, 0.305]
PolyBench-500	500	0.248 [0.227, 0.271]	0.363 [0.335, 0.391]	0.298 [0.277, 0.320]
SWE-bench Verified	500	0.158 [0.144, 0.174]	0.265 [0.244, 0.287]	0.155 [0.142, 0.169]
Pooled	1500	0.214 [0.203, 0.226]	0.328 [0.314, 0.343]	0.246 [0.235, 0.257]

swept as separate cells at $B=8000$ (Table 9). Removing the relatedness bonus or disabling adaptive stopping ($\tau=0$) moves pooled recall and precision by under one point — both mechanisms are recall-neutral at this budget, and adaptive stopping earns its keep on latency rather than quality. Removing the scoring-side lexical bonus ($\epsilon=0$; the BM25 top-1 discovery contribution and lexical Similarity edges remain active, so this is not a pure graph-only cell) is likewise nearly headline-neutral (0.923 vs. 0.919) but drops *retrieval* substantially (0.308 vs. 0.352 nontrivial recall, Section 6.7) — headline recall hides exactly the component interaction that matters. On the nontrivial metric the ablations reorder: disabling adaptive stopping ($\tau=0$) *raises* retrieval to 0.370 (stopping trades ≈ 1.8 points of nontrivial recall for its token savings), the relatedness bonus is neutral there too (0.349), and Boltzmann is comparable (0.356) — retrieval differences the headline column entirely hides. The Boltzmann objective raises headline recall slightly (0.928) while collapsing precision to 0.084: it fills the budget indiscriminately and is dominated by the default objective.

6.6 Budget Curve

To support claims about budget efficiency, the same protocol is repeated under EGO scoring (deployed default) at $B \in \{0, 8k, 16k, 32k, 64k, 128k, -1\}$ on the full 1500-instance test set. $B=0$ is a recall-floor sanity bound (must report exactly 0 at the sweep commit); $B=-1$ is *not* an unlimited-budget recall ceiling but the unbounded-token-cap variant of the same heuristic: the token knapsack relaxes to a 10M-token soft ceiling while the stopping rule, relevance filters, and per-source edge cap all remain active, so the -1 row is the heuristic-bounded asymptote of the deployed selector. The curve is reported in Table 10.

Figure 1 overlays the pooled budget curves of the three methods; the per-benchmark breakdowns with CIs remain in Tables 10–12.

For comparison, Table 11 reports the Aider repo-map fair-input baseline. The consolidated rerun measures Aider at the headline budget $B=8000$ only; the full Aider budget grid is inherited from the previous snapshot as indicative context rather than rerun data, since re-running the five non-headline Aider budgets (~ 19 hours of helper compute) is not justified for a baseline

Table 9: Operational ablations at $B=8000$, pooled over 1500 instances, intent-to-treat. Each row is a separate sweep cell toggling one mechanism via its clean-run flag (DIFFCTX_EGO_LEXICAL_EPS=0, DIFFCTX_OBJECTIVE=boltzmann, DIFFCTX_RELATEDNESS_BONUS=0, --tau 0) against the deployed EGO default. The nontrivial-recall column is the ContextBench Verified retrieval subset (Section 6.7; $n=213\text{--}215$ *ok* instances per cell). On the two headline metrics the default is Pareto-dominated by the no-bonus and $\tau=0$ rows; the nontrivial column shows $\tau=0$ genuinely leads retrieval as well — adaptive stopping and the relatedness bonus survive on latency and engineering grounds, not quality, and are candidates for re-examination in the next calibration cycle.

Variant	Pooled recall	Pooled precision	Nontrivial recall (CB)
EGO (deployed default)	0.919 [0.908, 0.929]	0.214 [0.203, 0.226]	0.352
no lexical bonus ($\epsilon=0$)	0.923 [0.912, 0.933]	0.234	0.308
no relatedness bonus	0.927 [0.916, 0.936]	0.217	0.349
no adaptive stopping ($\tau=0$)	0.929 [0.919, 0.939]	0.216	0.370
Boltzmann objective	0.928 [0.919, 0.938]	0.084	0.356

Table 10: Budget curve: pooled file recall under EGO mode (deployed default) across the budget grid, on the full 1500-instance test set. *Mean used tokens* is the per-instance mean of selected-fragment token counts under `tiktoken o200k_base`; the gap to the nominal budget reflects the lazy-greedy stopping rule rather than the budget being the binding constraint.

Budget B	n	Mean recall [95% CI]	Mean used tokens
0	1500	0.000 [0.000, 0.000]	0
8,000	1500	0.919 [0.908, 0.929]	4,915
16,000	1500	0.930 [0.920, 0.940]	9,378
32,000	1500	0.935 [0.925, 0.945]	16,087
64,000	1500	0.940 [0.930, 0.949]	24,222
128,000	1500	0.941 [0.931, 0.951]	32,533
$-1 (\infty)$	1500	0.948 [0.939, 0.957]	120,244

whose headline-budget comparison is decisive. The previous snapshot showed a $16\times$ budget gap in `diffctx`’s favor (Aider at $B=128,000$ below `diffctx-EGO` at $B=8000$).

The same sweep for the external BM25 baseline (Table 12) sharpens a headline finding of the rerun: unlike the previous snapshot, in which whole-file lexical packing overtook `diffctx-EGO` beyond $\sim 32k$ tokens, the rerun’s `diffctx-EGO` stays ahead at *every* measured budget (0.935 vs. 0.846 at 32k; 0.941 vs. 0.921 at 128k), with the gap widest in the tight-budget regime motivating the problem (Section 3, cost model and “lost in the middle” [19]). The within-framework internal-BM25 cell (Table 8) separates the lexical-signal question from the whole-file-packing question: the framework’s fragment-level machinery lifts BM25 from whole-file packing (0.564 pooled at $B=8000$) to the strongest headline-recall mode (0.942 at 2.9k mean used tokens) — most of the external gap at tight budgets is packing granularity, not the relevance signal.

6.7 Trivial Retention vs. Context Retrieval

The evaluation protocol feeds the gold patch itself to the selector as the input diff, so every method trivially includes the diff’s own files (changed-file retention is ≥ 0.99 for all `diffctx` modes). Headline file recall therefore conflates two different abilities: *retaining* the files the

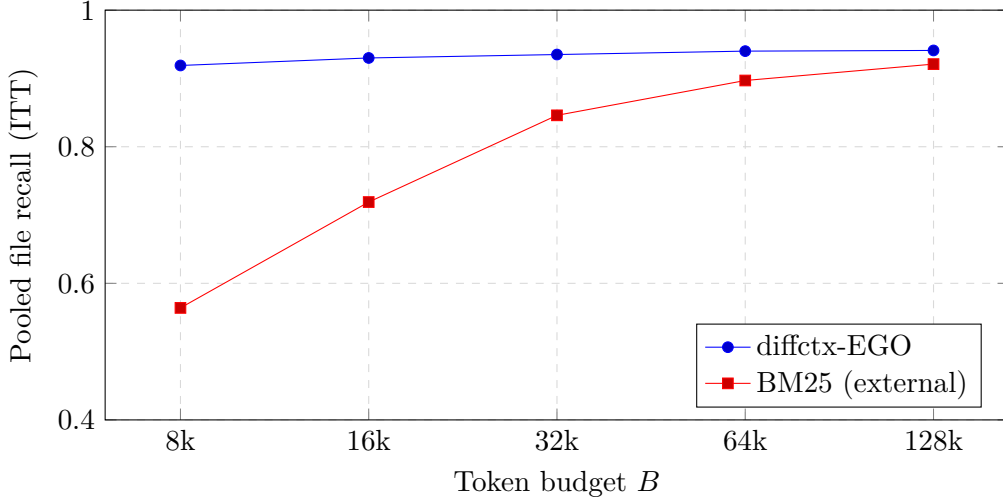


Figure 1: Pooled file recall vs. token budget (intent-to-treat, $n=1500$), consolidated rerun. diffctx-EGO is nearly flat—0.919 at $B=8000$ already—while whole-file BM25 packing needs the full 128k budget to approach it and never crosses it in the measured range. The Aider fair-input rerun point at $B=8000$ is 0.493 pooled (Table 11); its non-headline budgets are previous-snapshot indicative data.

Table 11: Aider repo-map fair-input baseline at the same instance-level inputs, intent-to-treat over $n=500$ per test set ($n=1500$ pooled). The $B=8000$ row is the consolidated rerun (measured with the helper-process stderr-deadlock fix that invalidated a fraction of the previous snapshot’s timeout rows); the greyed non-headline budgets are inherited from the previous snapshot as indicative context only and pair against the superseded diffctx run. $B=0$ is the recall-floor sanity bound; $B=-1$ is excluded as Aider does not accept an unlimited-budget directive.

Budget B	n	ContextBench Verified	PolyBench-500	SWE-bench Verified	Pooled
0	1500	0.000	0.000	0.000	0.000
8,000 (rerun)	1500	0.468 [0.432, 0.505]	0.316 [0.280, 0.353]	0.694 [0.655, 0.733]	0.493 [0.470, 0.515]
<i>previous snapshot (indicative only):</i>					
16,000	1500	0.509	0.356	0.766	0.544
32,000	1500	0.601	0.401	0.839	0.614
64,000	1500	0.662	0.455	0.897	0.671
128,000	1500	0.714	0.527	0.916	0.719

reader already knows about, and *retrieving* related context files the diff does not name. The harness records the decomposition per instance: `gold_to_changed_ratio` (1.0 means gold coincides with the diff) and `nontrivial_file_recall` (recall restricted to gold files *outside* the diff).

Applying the decomposition to the three test sets is clarifying. On SWE-bench Verified and PolyBench-500, *no instance* has gold files outside the diff — their gold sets derive from the patch, so their recall columns measure retention plus budget compliance. A baseline that simply echoes the diff’s files is near-optimal on them *at fragment granularity*; the whole-file `patch_files` floor scores only 0.549/0.741 there (Section 6) because whole files overflow the budget — the gap to diffctx’s 0.998/0.942 is fragment-level packing, not retrieval. ContextBench Verified is the only test set with human-annotated context beyond the diff: 213 of its 500 instances carry nontrivial gold (573 files total). Table 13 reports the retrieval-only comparison on that subset.

Three readings follow. First, all methods clear the trivial floors by an order of magnitude — context retrieval is real — and the external Aider repo-map lands within measurement noise of the deployed EGO mode (0.354 vs. 0.352): despite the wide headline gap (Table 7, driven almost

Table 12: Budget curve for the external BM25-over-patch-identifiers baseline, same protocol (intent-to-treat, $n=500$ per test set), consolidated rerun. BM25 does not support $B=-1$ (it packs whole files in rank order to the budget).

Budget B	n	ContextBench Verified	PolyBench-500	SWE-bench Verified	Pooled
0	1500	0.000	0.000	0.000	0.000
8,000	1500	0.485 [0.451, 0.519]	0.667 [0.634, 0.700]	0.540 [0.497, 0.582]	0.564 [0.543, 0.585]
16,000	1500	0.626 [0.594, 0.657]	0.775 [0.745, 0.803]	0.756 [0.719, 0.791]	0.719 [0.699, 0.737]
32,000	1500	0.738 [0.711, 0.765]	0.868 [0.845, 0.890]	0.931 [0.909, 0.951]	0.846 [0.831, 0.859]
64,000	1500	0.807 [0.783, 0.830]	0.916 [0.897, 0.933]	0.970 [0.955, 0.982]	0.897 [0.886, 0.909]
128,000	1500	0.851 [0.829, 0.871]	0.930 [0.913, 0.946]	0.981 [0.970, 0.991]	0.921 [0.911, 0.931]

entirely by Aider’s weaker changed-file retention), the two systems’ genuine retrieval ability is indistinguishable. Second, the lexical signal is currently the strongest single retrieval signal (0.402 for internal BM25): the deployed default EGO, selected by recall-oriented calibration, trails the plain lexical mode on the very ability the calibration intended to optimize. Third — and most consequential for the method — the signals are *complementary*, not redundant: EGO finds 41 nontrivial gold files that BM25 misses (BM25 finds 63 that EGO misses). A score-free *union fusion* — run both modes, emit the union of their selections — is directly runnable and realizes the combination: 0.473 nontrivial recall and 0.870 headline recall, both above every single mode, at precision 0.181. Its measured deduplicated token cost is mean 6,101 on the retrieval subset — inside the 8k budget for 91.5% of instances, since the two modes share core fragments (Table 13 caption); the residual 8.5% over-budget tail is handled by the sequential form — BM25 first (≈ 2.9 k mean), EGO on the per-instance remainder. The deployed EGO mode, which mixes graph and lexical signals with hand-set weights *inside* one scorer, lands *below* its own best component ($0.352 < 0.402$) — the classic signature of a miscalibrated mixture; the fusion result shows the combination value is real and bounds what per-edge-category weight calibration (Section 4.3) should recover *without* the fusion’s precision cost. The budget curve on the nontrivial subset also differs qualitatively from the headline: nontrivial recall for EGO grows from 0.352 at 8k to 0.503 at 32k, 0.581 at 128k, and 0.684 at the unbounded cap. That last number is a *universe-coverage* bound: even with no budget constraint, 31.6% of nontrivial gold is never selected — lost to discovery and relevance filtering, not to the budget (Section 4.2, universe construction). The 8k deficit therefore decomposes roughly half-and-half into files the pipeline never surfaces and files surfaced but not selected under budget.

6.8 Limitations of the Current Empirical Snapshot

All numbers in this section reflect the completed consolidated rerun at git tag `paper-v2` (v4 calibration, status-gated patch application, two-pass edge construction), with the sole exception of the greyed non-headline Aider budget rows. The following items are honest open questions about the snapshot:

- **Aider non-headline budgets are inherited, not rerun.** All $B=8000$ cells (including both Aider modes) are consolidated-rerun measurements; only the non-headline rows of the Aider budget curve (Table 11) remain previous-snapshot, greyed and marked indicative. The preregistered P1 and P3 confirmatory tests are resolved (both in diffctx’s favor); P2 is resolved as a significant omnibus with no pairwise separation on headline recall.
- **Two of three test sets measure retention, not retrieval.** SWE-bench Verified and PolyBench-500 carry no gold files outside the input diff (Section 6.7), so their recall columns are changed-file-retention measurements; genuine retrieval is measurable only on the 213-instance nontrivial subset of ContextBench Verified. The headline pooled recall is

Table 13: Retrieval-only comparison at $B=8000$: mean per-instance `nontrivial_file_recall` over the ContextBench Verified instances whose gold sets extend beyond the input diff. Convention: all mode numbers in this table are computed on the *common* ok-subset across the compared modes ($n=213$, 573 nontrivial gold files); per-cell ok-subsets differ by up to 3 instances and shift single-mode means by ≤ 0.005 (e.g. BM25 0.397 on its own $n=216$ subset vs. 0.402 here). The union-fusion rows are *runnable*, score-free combinations: run both modes and emit the union of their selected sets. The fused output also leads headline recall on this test set (0.870 vs. 0.846 single-best) at precision 0.181. Its *deduplicated* token cost, measured per instance by re-running both modes and tokenizing the union of unique fragments (`fusion_union` artifact), is mean 6,101 on this subset (p50 6,151, p90 7,816) — inside the 8,000 budget for 91.5% of instances, because the two selections share their core fragments; the pre-deduplication sum (mean 8,169, 49.8% over budget) is the naive-concatenation upper bound. The 8.5% of instances whose deduplicated union still exceeds the budget (p100 14,546) need the strictly compliant sequential variant (BM25 first, EGO on the per-instance remainder). The artifact’s live re-computation also reproduces the 0.473 nontrivial recall exactly. Trivial floors: the changed-files-only baseline scores 0.000 by construction; random packing scores 0.012.

Method	Nontrivial recall
Changed-files floor (<code>patch_files</code>)	0.000
Random packing floor	0.012
PPR	0.244
EGO without lexical scoring bonus ($\epsilon=0$)	0.308
EGO (deployed default)	0.352
Aider repo-map (fair-input, external)	0.354
BM25 (internal)	0.402
Aider oracle-mentioned (gold hinted; upper bound)	0.452
Union fusion EGO $\cup$ BM25 (runnable)	0.473
Union EGO $_{\epsilon=0}$ $\cup$ BM25	0.475

therefore dominated by retention. Future evaluation work should prioritize benchmarks with human-annotated context beyond the diff.

- **The deployed mixture is miscalibrated.** The EGO mode’s hand-set combination of graph and lexical signals scores below its own lexical component on nontrivial retrieval (0.352 vs. 0.402; oracle union 0.473). Per-edge-category weight calibration from inclusion-provenance data (Section 4.3) is the highest-leverage open item; the emission-side instrumentation for it (per-category edge counters in the run telemetry) has landed post-freeze.
- **PolyBench-500 fragment-level metrics are not reported.** PolyBench-500 ships a `modified_nodes` field encoding CST node paths (e.g., `file.java->program->class_declaration:Foo->method_declaration:bar`) rather than line ranges, and the current adapter does not yet resolve these symbolic paths to `(start_line, end_line)` tuples via a tree-sitter pass on the post-patch worktree. PolyBench-500 therefore contributes to file-level recall only; fragment-level and line-level metrics on this benchmark are deferred.
- **Residual non-ok rows.** Pooled ok% is 97.5%. The residue decomposes into: 5 deterministic PolyBench-500 patch-application failures — verified to be CRLF-encoded repository blobs vs. LF-normalized gold patches in the upstream dataset (a whitespace-tolerant apply accepts all five); per-instance 900s timeouts concentrated in the near-dense-graph repository class (the `mui/material-ui` family, where pre-cap edge emission reaches $\sim 2 \times 10^8$); and 4 clone failures. The former OOM-kill class is eliminated by the two-pass bounded

edge construction; the timeout class is the remaining engineering target, with per-category emission counters now in telemetry to attribute the blowup to a specific edge-builder family.

- **Per-future timeout.** The runner's per-future timeout does not enforce in single-worker mode because `concurrent.futures.as_completed` applies a bulk deadline rather than per-call. A small fraction of instances exceed the configured deadline; this increases wall-clock but does not corrupt results.

7 Discussion and Limitations

Language Coverage. The framework requires language-specific tooling for AST parsing and name resolution. Coverage follows the taxonomy of Appendix D (Table 16): structural parsing for ≈ 40 formats; 36 dedicated semantic edge-builder modules (a module may cover several languages) plus a separate config-edge family; structural fragmentation without semantic edges for the remainder; and empirical validation ($n \geq 10$ evaluation instances) for eight languages only (Python, TypeScript, JavaScript, Java, Go, C, Rust, C++). Everything outside that validated tier is a code-level claim, not an empirical one. Dynamic languages present challenges due to incomplete static analysis of dynamic dispatch.

Symbol Reference vs. Data-Flow. Our “symbol reference” edges use lexical name matching within function scope as a heuristic. This is **not** true data-flow analysis, which requires CFG construction and reaching definitions computation—infeasible for incomplete code in git working trees. We explicitly acknowledge this limitation.

Forward-Backward PPR Sensitivity. The implementation runs PPR on both the original graph and its transpose and linearly blends the two scores (Equation 11); for change-understanding the default is backward-leaning ($\rho=0.4$). The blend weight ρ has not been re-calibrated per language or per benchmark; sensitivity to ρ is a natural extension and is recorded as future work. A structural caveat applies to any EGO-vs-PPR conclusion: both modes score the *discovery-restricted* fragment universe (Section 4.2), so PPR’s “global propagation” is global only within that subgraph, and the near-tie between the modes partly reflects the shared discovery bottleneck rather than a pure statement about the radius of relevance.

Hub Node Dominance. Utility classes with high in-degree (Logger, Config, Utils) can accumulate excessive PPR mass. Our implementation applies two mechanisms (Heuristics Annex): IDF-style damping $w'_{uv} = w_{uv} / \log(1 + \text{in_degree}(v))$ for nodes exceeding the 95th percentile in-degree, and an out-fan mechanism damping **Semantic** edges from source nodes that reference three or more distinct files ($w/\sqrt{\text{fan}}$). The two do not compose the way the motivation suggests: the in-degree mechanism *exempts* the **Semantic**, **Structural**, and **TestEdge** categories, yet canonical utility hubs accumulate their mass precisely through **Semantic** call/import edges—so the canonical hub scenario is untouched by in-degree damping and covered only partially by the out-fan rule. The exemption encodes a judgment that structural edges are load-bearing and should not be degree-penalized; whether the net configuration actually suppresses hub dominance is an open empirical question, and a hub-suppression on/off ablation is part of the consolidated sweep.

Configuration Files. Non-code files (YAML, JSON, TOML) lack structural dependencies. For these, lexical signals must dominate, with higher BM25 weights (0.4–0.5).

Parameter Sensitivity. While we provide operational ranges for parameters (α , τ , edge weights), optimal values are repository-dependent. We recommend calibration on 20–50 historical commits via grid search.

8 Conclusion

We have presented `diffctx`, a framework for diff-aware code context selection that formulates the problem as budgeted utility maximization over multi-resolution fragments on a typed dependency graph. The deployed system maximizes a monotone submodular concept-coverage objective with lazy-greedy budgeted selection and engineering heuristics, including an adaptive stopping rule that carries an additive loss certificate (Proposition 1). A pluggable scoring abstraction (EGO, PPR, BM25) allows the same selection algorithm to be instantiated with different relevance signals without changing the optimization machinery; bounded ego-network expansion (EGO) is the deployed default, though the consolidated rerun’s ablation (Section 6) shows the three modes trade off along different axes and the graph-lexical mixture is not yet calibrated to dominate its components. We do not claim a new approximation algorithm; we use known optimization structure to make context selection explicit, analyzable, and extensible.

Key contributions: (1) an explicit “algorithm $\times$ constraint $\times$ guarantee” map (Table 1) that separates the deployed heuristic from analyzable variants of the framework, with the submodular concept-coverage objective (Section 4.5) as the deployed default and the modular Boltzmann variant as opt-in; (2) a typed-edge dependency graph with per-category treatment (hub-suppression exemption for semantic, structural, and test edges); (3) empirical results from the consolidated rerun (Section 6) showing pooled file recall 0.919 [95% CI 0.908, 0.929] at $B=8000$ on 1500 multi-language instances under the deployed-default EGO scoring mode (per-benchmark recall: SWE-bench Verified 0.998, PolyBench-500 0.942, ContextBench Verified 0.816), a within-framework scoring-mode ablation in which no mode dominates (internal BM25 leads headline recall at 0.942 pooled, PPR leads precision at 0.328 pooled), and a budget curve $B \in \{0, 8k, 16k, 32k, 64k, 128k, -1\}$ showing recall 0.919 at $B=8000$ saturating to a heuristic-bounded asymptote of 0.948; and (4) a retention/retrieval decomposition of the evaluation metric (Section 6.7) showing that headline recall is dominated by trivially retaining the diff’s own files, that on the genuine-retrieval subset the lexical signal beats the deployed graph-lexical mixture (0.402 vs. 0.352 nontrivial recall), and that graph and lexical hits are complementary — a runnable union of the two modes’ selections reaches 0.473 nontrivial and 0.870 headline recall at 6.1k mean deduplicated tokens — localizing the framework’s open problem in mixture calibration rather than in either signal. Paired both-OK comparisons against external baselines all favor `diffctx` decisively: $\Delta = +0.371$ [$+0.350, +0.392$] over whole-file BM25 packing, $+0.423$ [$+0.399, +0.447$] over Aider repo-map fair-input, and $+0.410$ [$+0.385, +0.434$] over the Aider oracle-mentioned upper bound (permutation $p = 10^{-5}$ throughout, $n_{\text{both-OK}} \geq 1421$); in this rerun external BM25 no longer overtakes `diffctx`-EGO at any measured budget, removing the previous snapshot’s tight-budget-only qualifier.

8.1 Future Work

Additional behavioral proxies. The current evaluation uses file-level recall against an annotated golden set. Three complementary proxies are natural extensions:

- **Dependency coverage recall.** For a diff Δ , define $\text{Coverage}(C, \Delta) = |\{d \in \text{deps}(\Delta) : \exists f \in C, d \text{ resolves in } f\}| / |\text{deps}(\Delta)|$, where $\text{deps}(\Delta)$ is the set of definitions of modified symbols, call targets, callers, types in signatures, and covering tests. This complements file recall by measuring whether the selected context resolves the symbols that appear in the diff [25].
- **Co-change recall.** For commits where files A and B change together, input only A ’s diff and measure whether B appears in C . Co-change has a 30–40% incidental-coupling false-positive rate [11]; the proxy is meaningful only after filtering to semantic modifications (≥ 5 substantive lines).

- **Fault-localization accuracy.** Given (Δ, C) , an LLM-based or human evaluator ranks suspicious lines for root-cause identification, scored by Mean Reciprocal Rank and Precision@ k [25].

Edge-weight calibration on labeled corpora. Section 4.3 fixes w_t at author-selected priors informed by the code-navigation literature. A larger labeled corpus (e.g., a sweep across all six benchmarks combined) would enable the full $|T|$ -dimensional calibration of w_t , with the formulation already laid out there. Richer parameterizations (per-edge MLP over edge-context features) become tractable once labeled data scales further.

Blend sensitivity, memory, and dynamic dispatch. Forward-backward PPR blending is already deployed (Equation 11, $\rho = 0.4$); per-language and per-benchmark sensitivity to ρ has not been studied and is a direct extension. The Go out-of-memory class has been fixed by bounded two-pass edge construction: 34 of 40 Go instances now complete (ok-only recall 0.921, Table 6); the residual 6 are per-instance timeouts on near-dense-graph repositories (Section 6.8). Per-language semantic resolvers (e.g. for interface-method-set dispatch and metaprogramming-heavy code) remain valuable for the residual gap between dynamic and statically-typed languages on completed instances.

Acknowledgments

The reference implementation lives at <https://github.com/nikolay-e/diffctx>.

A Parameter Guidelines

Parameter	Range	Selection Guidance
α (damping)	0.50–0.65	Start at 0.60; decrease for over-selection of distant code
τ (threshold)	0.04–0.16 (grid)	Deployed calibrated default 0.12; 0.08 is an uncalibrated heuristic for narrowly-scoped fixes
Budget B	4k–20k tokens	8k is the evaluated standard; the CLI default at the rerun commit was 4096 (<code>DEFAULT_BUDGET_TOKENS</code> , since replaced by an auto-budget mode); scale with diff complexity
Call weight	0.55–0.70	Matches <code>config/weights.rs</code> (Section 4.3); tune via cross-validation on historical commits
Lexical weight	0.1–0.2	Reduce if high false positive rate
Overhead	15–20 tokens	Measure actual serialization cost per fragment

Table 14: Operational parameter ranges with selection heuristics.

Implementation parameters not modelled in Section 3–4.6. The reference implementation exposes additional parameters that govern second-order behaviors not formalized in the body: a relevance-augmentation weight `UtilityConfig.eta` (0.20) blending raw PPR scores into match strength; a structural-bonus weight `UtilityConfig.structural_bonus_weight` (0.10) adding a small bonus when a fragment has any matched need; an r_{cap} calibration constant `UtilityConfig.r_cap_sigma` (2.0, in standard deviations) controlling the soft cap on per-fragment relevance contribution; a hunk-proximity decay system (`FilteringConfig.proximity_half_decay` = 50 lines, `FilteringConfig.definition_proximity_half_decay` = 5 lines, `UtilityConfig.proximity_decay` = 0.30) that boosts fragments physically near diff hunks; BM25 hyperparameters `Bm25Config.idf_smoothing` (0.5) and `Bm25Config.min_query_token_length` (3), with the average document length computed from the corpus; the Boltzmann calibration loop’s bisection bounds `BoltzmannConfig.beta_lo` (10^{-6}), `BoltzmannConfig.beta_hi` (1.0), iteration cap `BoltzmannConfig.bisect_iters` (24), and convergence tolerance `BoltzmannConfig.calibration_tolerance` (5% of budget); and the rescue-phase parameters `RescueConfig.budget_fraction` (0.05) and `RescueConfig.min_score_percentile` (0.80). These parameters are calibrated empirically on representative corpora and held fixed across modes; their role is operational rather than theoretical, so we do not model them in the body but document them here for reproducibility.

B Algorithmic Complexity

Component	Time	Space	Practical Scale
AST parsing	$O(n)$	$O(n)$	Tested to $\sim 500\text{k}$ LOC
Semantic/structural edge builders	$O(n \log n)$	$O(n + m)$	Tested to $\sim 200\text{k}$ LOC
Similarity (TF-IDF) edges	$O(n \cdot \bar{t})$, $\bar{t}$ = tokens per fragment	$O(n + m)$	—
History (co-change) edges	$O(h)$, h = git-log entries	$O(m)$	—
PPR (forward-push)	$O(\text{pushes})$, cap $\min(100n, 2 \times 10^6)$	$O(n + m)$	Tested to $\sim 50\text{k}$ nodes
Lazy greedy	$O(\mathcal{V} \log \mathcal{V})$ typical; $O(\mathcal{V} ^2)$ re-computations worst case	$O(n)$	Tested to $\sim 5\text{k}$ fragments
Post-passes	$O(C \cdot \bar{d} + n)$	$O(n)$	Bounded by remaining budget

Table 15: Complexity analysis where n = fragments, m = edges, $\bar{d}$ = average degree. PPR uses a forward-push (local-push) solver with a global push cap, not power iteration, so its cost is measured in pushes rather than iteration count. Practical scale limits are estimates requiring validation; performance depends on repository structure and hardware. Empirical run-time and memory telemetry (p50/p90/p99 per-instance latency, peak RSS, node/edge counters, per-phase shares, timeout/OOM rates) is collected per instance by the harness and will be reported from the consolidated rerun’s logs.

C Heuristics Annex

The following tactics are useful in practice but lack strong theoretical or empirical grounding. They should be evaluated via ablation before deployment.

Relatedness (Diversity) Bonus. Fragments with relevance above a small threshold (≥ 0.03 , `NeedsConfig.min_rel_for_bonus`) receive an additional marginal gain $\text{rel} \times 0.25 \times \text{unsatisfied}(C)$, where $\text{unsatisfied}(C) = \max(0, 1 - \sum_z \text{cov}_z(C)/|Z|)$ is the fraction of concept mass not yet covered by the selected set. This keeps structurally related fragments in play early while fading the bonus as coverage saturates. Separately, a modular structural bonus of $0.10 \cdot r_{\text{norm}}$ applies to fragments matching at least one need (`UtilityConfig.structural_bonus_weight`). **Warning:** the diversity term is path-dependent (its increment depends on coverage at selection time) and falls outside the strict set-function form, so worst-case approximation guarantees do not apply to the deployed configuration; its increment is pointwise non-increasing in C , which is the property Proposition 1 needs. Setting `DIFFCTX_RELATEDNESS_BONUS=0` disables it. An ablation with vs. without the bonus is still required.

Hub Suppression. Apply IDF-style penalty to edges into high in-degree nodes: $w'_{uv} = w_{uv} / \log(1 + \text{in_degree}(v))$ when $\text{in_degree}(v)$ exceeds the 95th percentile; edges in the exempt categories (`Semantic`, `Structural`, `TestEdge`) are not damped by this in-degree mechanism. A second, out-fan mechanism damps `Semantic` edges from source nodes referencing ≥ 3 distinct files: $w' = w / \sqrt{\text{fan}}$, targeting broad-capture import hubs. Both are implemented globally during graph construction, before the per-source top- K cap.

Backward Edge Weighting. Weight caller $\rightarrow$ callee edges at a per-language `reverse_factor` of $0.35\text{--}0.70\times$ (most languages 0.40 ; `config/weights.rs`) of callee $\rightarrow$ caller edges to prioritize impact analysis over dependency tracing.

Test File Handling. Test-to-code edges are static: direct links (test imports/references the unit) carry weight 0.60 , naming-convention links 0.50 , and reverse (code $\rightarrow$ test) links 0.30 (`config/weights.rs`). The system does not execute tests; upweighting tests that fail after Δ (e.g. to 0.9) requires runtime signal and is future work.

Signature Variants. Selection is fragment-granular, so partial function bodies never appear; instead, every full semantic unit gets a signature-only companion fragment (bracket-balanced signature extent, falling back to the first two lines; `signatures.rs`) that serves as its compact matroid representative when the full body does not fit.

Post-Passes. After the main greedy loop terminates (either by exhausting the budget or by adaptive stopping, Section 4.6), three ordered post-passes run (`postpass.rs`, `pipeline.rs`): (i) a *coherence pass* (`coherence_post_pass`) scans the selected set for dangling references—selected fragments whose semantic-neighbor definitions are absent from the context—and pulls in the missing definitions against the full remaining budget; (ii) a *rescue pass* (`rescue_nontrivial_context`) spends a reserved fraction (5% of B , `RescueConfig.budget_fraction`) on high-relevance fragments from files not yet represented, gated by a percentile threshold on relevance (`RescueConfig.min_score_percentile = 0.80`) so it does not flood the context with low-confidence rescues; (iii) a *changed-files pass* (`ensure_changed_files_represented`) ensures, budget permitting, that every file touched by Δ appears at least as a compact representative—adding without any relevance gate (it synthesizes a whole-file fragment if the file has none), but strictly within the remaining budget:

when no representative of a changed file fits, the file stays unrepresented and the shortfall surfaces as changed-file retention < 1 rather than as a budget overrun. All three passes respect $\text{cost}(C) \leq B$ and pairwise-overlap feasibility; for a monotone objective they never decrease utility below the greedy value, but they relax the strict density-greedy ordering, and the rescue and changed-files passes draw candidates from the pre-filter fragment universe, outside the scope of Proposition 1’s certificate (the coherence pass draws from the post-filter universe and stays within it).

D Language Support

Two independent layers make up language support, and they are counted separately because neither is a subset of the other. *Structural parsing* (fragmentation): 39 registered tree-sitter grammar keys (38 distinct grammars; JSX aliases the JavaScript grammar) plus a regex-structural TOML splitter and a heading-based Markdown splitter — about 40 structurally parsed formats. *Semantic edge extraction*: 36 dedicated edge-builder *modules* (excluding one generic cross-file identifier linker), where a single module may cover several languages (one JVM builder serves Java/Kotlin/Scala/Groovy; one C-family builder serves C/C++/Objective-C; the JavaScript builder also serves TypeScript), so the tier rows below enumerate *languages* and legitimately exceed 36. Three builders (LaTeX, Perl, Nim) operate on generically fragmented text without a dedicated parser — edge building keys off file extension and raw text, not the parse tree. A separate family of six *config* edge builders (Dockerfile, Kubernetes, Helm, CI/CD pipelines, build systems, generic config→code) emits the **Config/ConfigGeneric** categories of Table 3; the domain builders listed in Tier 2 (Ansible, dbt, OpenAPI, Cargo) sit on generic serialization formats at the parsing layer but re-parse raw text with dedicated extractors and emit **Semantic** edges. Coverage is uneven: a handful of languages have been exercised extensively in the evaluation; the rest are code-level claims. Table 16 groups by tier of empirical exercise.

Tier	Languages / formats
Tier 1 — empirically validated ($n \geq 10$ in evaluation)	Python ($n=891$), TypeScript ($n=191$), JavaScript ($n=185$), Java ($n=140$), Go ($n=40$), C ($n=23$), Rust ($n=20$), C++ ($n=10$)
Tier 2 — semantic edge builders, no large-scale empirical validation	C#, F#, VB, Kotlin, Ruby, PHP, Swift, Scala (general-purpose); Haskell, Elixir, Erlang, OCaml, Clojure, Julia (functional); Lua, R, Perl, Bash/Zsh/Fish/Ksh, Dart, Groovy, Objective-C, Zig, Nim (scripting/systems); CSS/SCSS/LESS, GraphQL, HCL/Terraform, Nix, LaTeX (markup/infra); Cargo, Bazel, dbt, OpenAPI, Prisma, Protobuf, SQL, Ansible (domain; Semantic edges over generic serialization formats)
Config edge builders (Config/ConfigGeneric categories)	Dockerfile/Compose, Kubernetes manifests, Helm charts, CI/CD pipelines, build-system files, generic config→code references
Tier 3 — parser only, structural fragmentation without semantic edges	YAML, JSON, TOML, Svelte, HTML, CMake, Make (bare files; CMake/Make cross-references are reachable via the build-system config builder)
Generic fallback	Markdown (structural heading splitter), plain text, dotfiles, unparsed sources (paragraph or statement fragmentation)

Table 16: Tiered coverage of the reference implementation. Tier 1 has both edge builders and per-language recall numbers in Table 6. Tier 2 lists *languages* covered by the 36 dedicated semantic edge-builder modules (a module may serve several languages, so rows exceed 36 by design); its inclusion is a code-level claim, not an empirical one. The config-builder family is a separate edge layer, not a language tier. Tier 3 supplies parsing only. Only Tier 1 carries empirical evidence.

E Symbol-to-Code Map

This appendix records the correspondence between symbols in the body of the paper and their identifiers in the reference Rust implementation. The map is the durable contract between the formalism and the code: a symbol that appears in the paper must resolve to a named field, function, or module in the codebase, and any drift surfaces here. Identifiers and counts reflect the rerun snapshot at git tag `paper-v2`; later commits replaced the fixed 4096 CLI default (`DEFAULT_BUDGET_TOKENS`) with an auto-budget mode and added an `Excerpt` fragment kind (22nd, not counted in Table 2). Paths are relative to `diffctx/src/`.

Paper symbol	Meaning	Code identifier	Source
α	PPR damping factor	<code>PPRConfig.alpha</code>	<code>config/limits.rs</code>
ρ	PPR forward-blend weight (Eq. 11)	<code>PPRConfig.forward_blend</code>	<code>config/limits.rs</code>
β	Boltzmann inverse temperature	<code>BoltzmannConfig.beta_lo/beta_hi</code> (bisection bounds)	<code>config/selection.rs</code>
β_{core}	Core edit budget fraction	<code>SelectionConfig.core_budget_fraction</code>	<code>config/selection.rs</code>
γ	Ego-network per-hop decay (Section 4.2.2)	<code>EgoScoringConfig.per_hop_decay</code>	<code>config/scoring.rs</code>
τ	Adaptive stopping threshold	<code>DEFAULT_STOPPING_THRESHOLD</code> (CLI <code>--tau</code>)	<code>config/limits.rs</code>
L	Ego-network expansion radius	<code>EgoGraphScoring.max_depth</code>	<code>scoring.rs</code>
k_1, b	BM25 saturation, length norm	<code>Bm25Config.k1, Bm25Config.b</code>	<code>config/bm25.rs</code>
w_t	Per-edge-category weight	<code>CategoryWeights.<variant></code>	<code>config/category_weights.rs</code>
$I(f)$	File-importance prior	<code>utility::importance::compute_file_importance</code>	<code>utility/importance.rs</code>
ω_f	Per-fragment importance scaling	<code>compute_seed_weights</code>	<code>core.rs</code>
E_0	Core edit set	<code>identify_core_fragments</code>	<code>core.rs</code>
F	Fragment universe	output of fragment-extraction phase	<code>fragmentation.rs, parsers/</code>
<i>Pipeline phases:</i>			
File discovery	Candidate file assembly	<code>candidate_files::collect_candidate_files</code>	<code>candidate_files.rs</code>
Fragment extraction	Tree-sitter fragmentation	<code>fragmentation::process_files_for_fragments</code>	<code>fragmentation.rs</code>
Symbol-level discovery	Seed expansion via rare idents	<code>discovery::*Discovery</code>	<code>discovery.rs</code>
Scoring	Pluggable $\hat{w}(f, \Delta)$	<code>PPRScoring, EgoGraphScoring, BM25Scoring</code>	<code>scoring.rs</code>
Filtering	Relevance/proximity filters	<code>filtering::*</code>	<code>filtering.rs</code>
Selection	Lazy greedy / Boltzmann	<code>select::lazy_greedy_select</code>	<code>select.rs</code>
Coherence post-pass	Rescue dangling references	<code>postpass::coherence_post_pass</code>	<code>postpass.rs</code>

Table 17: Symbol-to-code map. Paper symbols on the left correspond to the named code identifiers on the right, located in the listed source file under `diffctx/src/`. Implementation-only parameters are documented inline in Appendix A and are not duplicated here.

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